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Synthesis of pyrrole-ferrocene ensembles and their rearrangement into 2-(ferrocenylmethyl)-1,2-dihydro-3*H*-pyrrol-3-ones

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ABSTRACT

The lithiation of (ferrocenylmethoxy)allene followed by sequential reaction with isothiocyanate and S-alkylation of the adduct gives the corresponding alkyl buta-2,3-dienimidothioate (1-aza-1,3,4-triene), which in the presence of CuI or CuBr cyclizes into (ferrocenylmethoxy)/sulfanyl-substituted pyrrole ring. The process is carried out in one or two preparative steps depending on an alkylating agent. A novel unexpected rearrangement of the synthesized 3-(ferrocenylmethoxy)-2-sulfanyl-1*H*-pyrroles into 2-(ferrocenylmethyl)-2-sulfanyl-1,2-dihydro-3*H*-pyrrol-3-ones, proceeding in $CDCl_3$ at room temperature or in toluene under heating, is discovered. The transformation of the 3-(ferrocenylmethoxy)-1*H*-pyrroles into 1,2-dihydro-3*H*-pyrrol-3-ones is most likely the result of the formal [1,3]-O-to-C-rearrangements (acid-induced or thermal, respectively).

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1. Introduction

Both ferrocenes [1] and pyrroles [2] are among the well-known and the most popular classes of five-membered cyclic compounds, namely metallocenes and azoles, the interest to which is constantly increasing. The combination of a ferrocene and a pyrrole ring systems in a one molecule leads to the structures, even more interesting and promising, especially in the fields of organic synthesis, asymmetric catalysis, electrochemistry, materials science, biomedical research, and so on [3], than each of them separately. Some of pyrrole-ferrocene ensembles, in which ferrocene and pyrrole systems are bonded directly [4] or joined via a spacer, such as a methylene, carbonyl, amido, amidoalkyl, phosphine groups that separate the two ring systems [5], are described in the literature. Most of the known approaches to such ensembles are based on functionalization of preexisting pyrrole rings (unsubstituted or substituted) with ferrocene derivatives [3c–e,4a–c,5b,c]. The pyrrole-ferrocene ensembles were also accessed via construction of ferrocene ring system (starting from cyclopenta-1,3-dienyl-substituted pyrroles) [5e] and via construction of pyrrole ring (starting from suitable ferrocene derivatives) [4d,e]. However, methods for the synthesis of ferrocenyl-substituted pyrroles were rather limited.

(Alkylsulfanyl)pyrroles, displaying a broad spectrum of applications [6], are also attractive targets of great current interest as important organosulfur compounds as hetaryl sulfides [7] that are endowed with high reactivity, and properties valuable for pharmacology, optoelectronic and hi-tech materials. Among the known methods for the synthesis of multisubstituted 2-(alkylsulfanyl)–1*H*-pyrroles from acyclic precursors [8], the methodology using isothiocyanates both as a structural fragment of pyrrole core and supplier of the sulfur atom for sulfa-substituents, along with allenes or alkynes (as supplier of three carbon atoms for pyrrole ring and substituents for 3 and/or 5 positions), seems to be the most powerful, simple and perspective. Thus, the combination of different isothiocyanates with allenic or acetylenic carbanions, generated in situ from the corresponding allenes or alkynes, affords di- or trisubstituted 2-(alkylsulfanyl)pyrroles in good to excellent yields in one preparative step [9,10].

Nevertheless, pyrroles bearing ferrocenylalkoxy substituent or both ferrocene and alkylsulfanyl moieties, to the best of our knowledge, have not been hitherto described, but exactly from such pyrroles one might expect some novel interesting and unique properties aside from known very useful own properties of pyrroles, ferrocenes, and sulfides.

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Consequently, synthesis of ferrocenyl-substituted 2-(alkylsulfanyl)pyrroles is a challenging task, especially in the synthetic strategies based on available starting materials.

To initiate our study, we have synthesized the ferrocenic analogue of methoxyallene, i.e., (ferrocenylmethoxy)allene (**1**) [11], as a new building block and supplier of ferrocene functions for the one-pot synthesis of 3-(ferrocenylmethoxy)−2-sulfanyl-1*H*-pyrroles **2**.

2. Results and discussion

2.1. Synthesis of 3-(ferrocenylmethoxy)−2-sulfanyl-1H-pyrroles 2

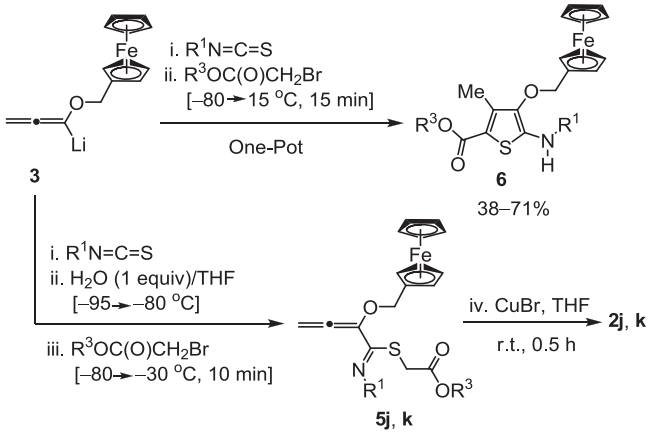
New ferrocene-substituted pyrroles **2a–i** were synthesized in one preparative step, according to our novel general approach to the pyrrole ring construction from isothiocyanates and allenes or alkynes [9,10], namely by the reaction of α-lithiated (ferrocenylmethoxy)allene (**3**) with isothiocyanate followed by the consecutive S-alkylation of adduct **4** and Cu(I)-catalyzed intramolecular [1,5]-cyclization of the forming alkyl allenylimidothioate (1-aza-1,3,4-triene **5**) (Table 1).

Lithiation of (ferrocenylmethoxy)allene (**1**) was carried out with *n*-BuLi in THF/hexane at −100 → −65 to −40 °C for 5 – 15 min. Temperature and time of the interaction of 1-(ferrocenylmethoxy)−1-lithioallene (**3**) with isothiocyanate (Table 1, Scheme, step ii) depends on structure of the latter. Thus, the complete addition of lithioallene **3** to methyl isothiocyanate occurs at −50 to −40 °C for 5 – 10 min, to 2-(vinyloxy)ethyl and phenyl isothiocyanates – at −45 to −40 °C for 10 min and at ca. −65 °C for 15 min, respectively. Addition of lithioallene **3** to secondary isothiocyanates, such as isopropyl, cyclopentyl, and cycloheptyl ones, was carried out at ca. −30 °C for 20 min. The forming lithium buta-2,3-dienimidothioates **4** were alkylated on the sulfur atom with methyl iodide, propargyl and benzyl bromides at −80 → −10 to −5 °C. The higher temperature for alkylation is not required because undesirable in this case isomerization (via [1,5]-H shift) of 1-aza-1,3,4-trienes **5** into conjugated 2-aza-1,3,5-trienes (precursors of 2,3-dihydropyridines) [9c,12] can take place (for N-Alk and N-cycloAlk derivatives). Addition of CuBr or CuI to the reaction mixture at low temperature (ca. −10 to −5 °C) followed by heating at 30–50 °C for 15–50 min led to intramolecular cyclization of 1-aza-1,3,4-triene **5** into pyrrole ring **2**.

The use of propargyl bromide as an alkylating agent (instead of methyl iodide) in the studied reaction (for example, with methyl isothiocyanate) resulted in 2-(prop−2-ynylsulfanyl)−1*H*-pyrrole **2 h** containing an additional reaction center (Table 1). Similar reaction with benzyl bromide led to the corresponding benzylsulfanyl derivative **2i**.

However, when alkyl bromoacetates were used as alkylating agents, modification of the reaction conditions was required. It was necessary in order to switch the 1-aza-1,3,4-trienes **5j,k** cyclization from the corresponding thiophenes **6** to pyrroles **2** (Scheme 1) [13]. This occurred when the 1-aza-1,3,4-trienes **5j,k** were preliminary isolated from the reaction mixture and then subjected to the action of CuBr (~9.0 mol%). Otherwise, i.e., when the reaction was carried out in one preparative step (as shown in Table 1 and Scheme 1), the formed *in situ* 1-aza-1,3,4-trienes **5**, bearing sulfanylacetate substituent, very rapidly cyclized to the corresponding alkyl 5-aminothiophene-2-carboxylates **6** [14].

In order to suppress the rapid cyclization of the 1-aza-1,3,4-trienes **5j,k** to corresponding thiophenes **6**, prior to the S-alkylation of the adducts **4j,k** (Table 1), one equivalent of water was added to the reaction mixture, and the treatment with alkyl bromoacetates was carried out at a lower temperature, namely at −80 to −30 °C for 10 mins (instead of −80 to 15 °C for 15 min) (Scheme 1).



Scheme 1. Suitable reaction conditions for switching the 1-aza-1,3,4-trienes **5j,k** cyclization from corresponding thiophenes **6** to the pyrroles **2j,k**.

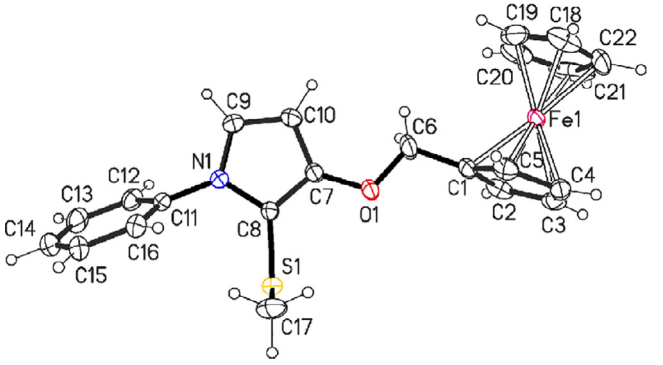


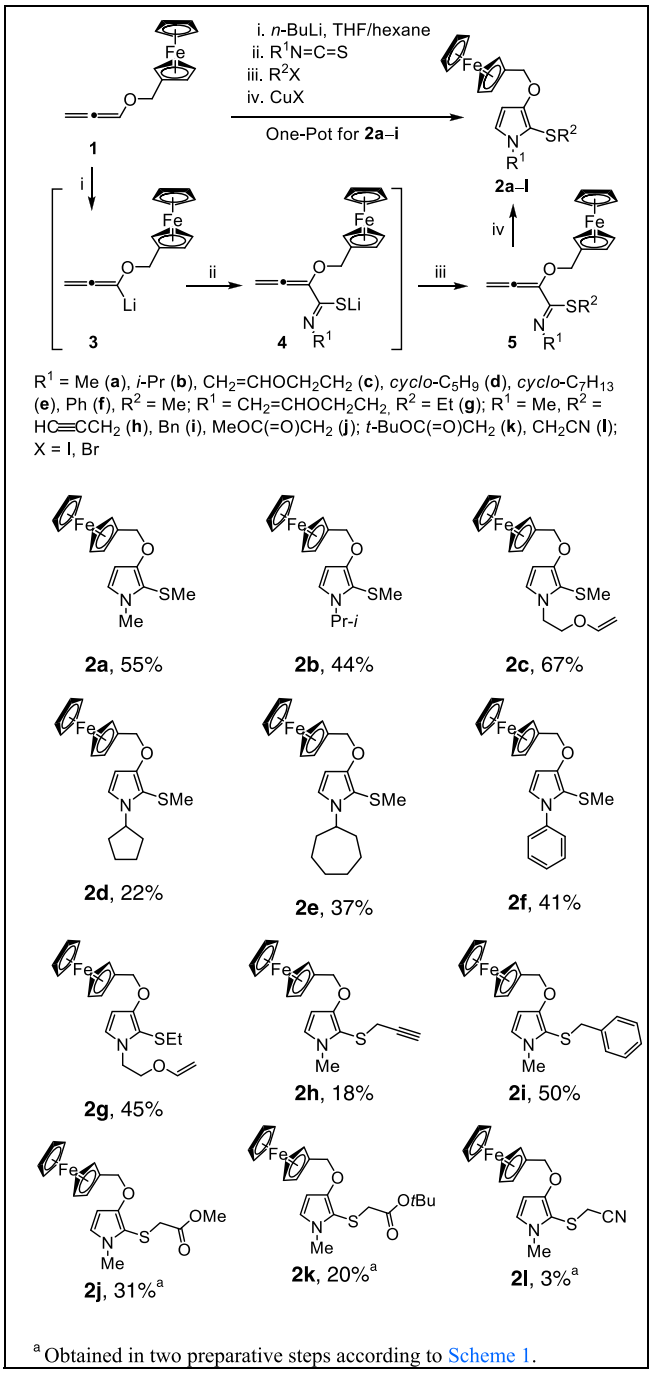
Fig. 1. Structure of molecule **2f** in crystal. Selected bond lengths (Å), angles (°), and torsion angles (°): C1–N6 1.484(6), O1–C6 1.428(6), O1–C7 1.368 (4), C8–S1 1.733(3), S1–C17 1.788(4), Fe1–C1–C6 123.4(3), C1–C6–O1 109.5(3), C6–O1–C7 114.2(3), C8–S1–C17 105.4(2), Fe1–C1–C6–O1 169.2(3), C1–C6–O1–C7 174.5 (3).

Similar reaction conditions were applied to the reaction of adduct **4a** with bromoacetonitrile (for the same reason, i.e., observed earlier extremely easy and fast intramolecular cyclization of generated in situ alkoxy-substituted 1-aza-1,3,4-trienes into corresponding tetrasubstituted thiophenes [15]). But the yield of isolated purified compound **2l** was only 3%.

Importantly, all these syntheses are scalable, and have been readily implemented on a gram scale. Non-optimized yields of pyrroles **2a–k** (after purification by column chromatography on neutral alumina) were 18−67% (calculated without taking into account the conversion of allene **1**, which was ~75−80%). All compounds, except for 2-(ethylsulfanyl)-, 2-(prop−2-ynylsulfanyl)-, and 2-(benzylsulfanyl)-substituted pyrroles **2 g, 2 h**, and **2i**, are crystalline from amber to yellow or orange substances; pyrroles **2 g, 2 h**, and **2i** are viscous liquids. The structures of the products have been assigned using ^{1}H, ^{13}C, and heteronuclear 2D (^{1}H,^{13}C HSQC, ^{1}H,^{13}C and ^{1}H,^{15}N HMBC) NMR and IR spectroscopy and mass-spectrometry. The molecular structure of pyrrole **2f** has been determined by X-ray diffraction (Fig. 1).

The structure of pyrrole **2f** was deposited by us under the number CCDC 1,015,374 and published in CSD Communication (Private Communication with refcode ECEJUU) [16]. In this article, we provide detailed experimental data, analysis of geometry and intermolecular interactions. Compound **2f** is a mono-substituted ferrocene with the two Cp-rings almost parallel [the angle between the least squares mean planes of the five-membered rings is 1.3(3) °]. The Fe−Cp interaction with the substituted C5-ring [distance from centroid Cg1 to atom Fe1 is 1.645(2) Å] is shorter than that with the unsubstituted Cp-ring [distance from centroid Cg2

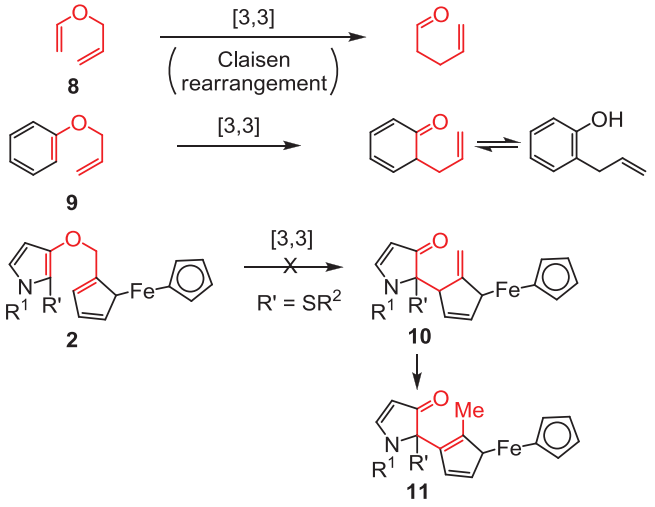
Table 1
Synthesis of 3-(ferrocenylmethoxy)−2-sulfanyl-1*H*-pyrroles **2** from (ferrocenylmethoxy)allene (**1**), isothiocyanates, and alkylating agents.



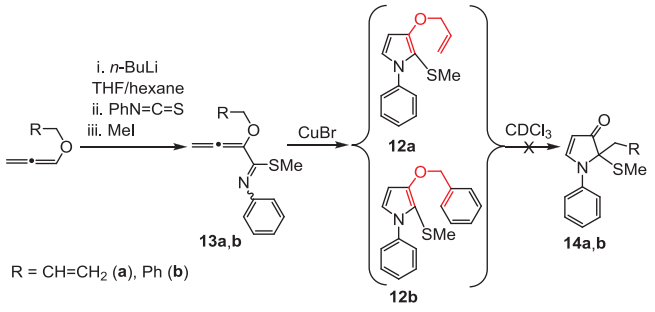
[a]Obtained in two preparative steps according to Scheme 1.

to atom Fe1 is 1.663(3) Å], the angle Cg1−Fe1−Cg2 is 176.7°(1). The Fe−C1−C6−O1 [169.2(3) °] torsion angle is close to antiperiplanar position for 2-(methylsulfanyl)−1-phenyl-1H-pyrrole substituent. The C1−C6 [1.484(6) Å] and O1−C6 [1.428(6) Å] bond lengths are typical for single bonds. The dihedral angle between the phenyl and pyrrole rings is 65.4°. The bond lengths and bond angles in the pyrrole ring are the same as the statistical means [17]. Infinite chains of molecules along a axis are formed in crystal with the help of intermolecular C−H···π interaction of H atoms of the C17−H13 and C10−H15 bonds with phenyl and substituted Cp rings correspondingly, the atom-to-plane distances being 2.72 and 2.96 Å.

Crystal data for pyrrole **2f** ($C_{22}H_{21}FeNOS$, M = 403.31 g/mol): monoclinic, space group Cc (no. 9), a = 5.9401(13) Å, b = 47.831(8) Å, c = 7.0601(15) Å, β = 108.950(7)°, V = 1897.2(7) Å^3, Z = 4, T = 200(2) K, μ(MoKα) = 0.915 mm^{-1}, Dcalc = 1.412 g/cm^3, 17,069 reflections measured (1.7 ° ≤ 2Θ ≤ 55.02 °), 4327 unique (R_{int} = 0.0545, R_{sigma} = 0.0493) which were used in all calculations. The final R1 was 0.0343 [for 3792 I > 2sigma(I)] and wR2 was 0.1008 (all data).



Scheme 2. Classical Claisen rearrangements.



Scheme 3. Synthesis of 3-allyloxy- and 3-benzyloxypyrroles **12**.

*2.2. Formal [1,3]-O-to-C-rearrangement of 3-(ferrocenylmethoxy)–2-sulfanyl-1H-pyrroles **2** into 2-(ferrocenylmethyl)–2-sulfanyl-1,2-dihydro-3H-pyrrol-3-ones **7***

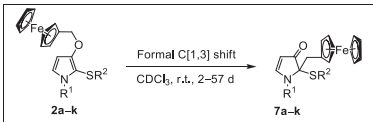
It was noticed that during storage of a solution of pyrroles **2a**–***k*** in $CDCl_3$ at room temperature, their color was changed in time from yellow-orange to dark-brown. Repeated NMR spectra of these samples recorded after 2 to 57 days (or 4.5 and 4 months for pyrroles **2j** and **2k**, respectively) showed that 3-(ferrocenylmethoxy)–2-sulfanyl-1*H*-pyrroles **2** partially or fully transformed into previously unknown 2-(ferrocenylmethyl)–2-sulfanyl-1,2-dihydro-3*H*-pyrrol-3-ones **7** (Table 2).

It is important to note, that although pyrroles **2** have a structural fragment (highlighted in red), which with a certain degree of conditionality can be attributed to allyl vinyl (**8**) or allyl aryl (**9**) ethers, unlike the latter [18], they are not exposed to classical Claisen rearrangement (under the above conditions) with the formation of 1,2-dihydro-3*H*-pyrrol-3-ones **10** and/or **11** (Scheme 2).

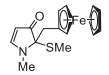
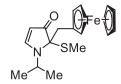
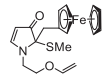
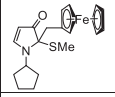
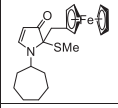
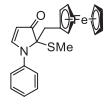
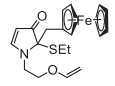
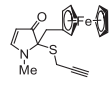
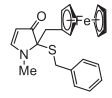
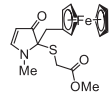
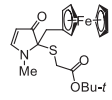
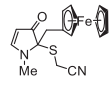
On the other hand, earlier unknown 3-allyloxy-2-(methylsulfanyl)–1-phenyl-1*H*-pyrrole (**12a**) and 3-benzyloxy-2-(methylsulfanyl)–1-phenyl-1*H*-pyrrole (**12b**), specially synthesized from allyloxy- and benzyloxyallenes, phenyl isothiocyanate and methyl iodide (via CuBr-catalyzed cyclization of 1-aza-1,3,4-trienes **13**), the most closest (by structure) analogues of pyrrole **2f** and allyl hetaryl ethers, under the similar conditions ($CDCl_3$, r.t., ~20 days) did not rearrange into the corresponding 1,2-dihydro-3*H*-pyrrol-3-ones **14** (Scheme 3), the ferrocenylmethoxy-substituted analogue **2f** being rearranged quantitatively (Table 2).

Products of the Claisen rearrangement of pyrroles **12**, i.e., 1,2-dihydro-3*H*-pyrrol-3-ones **14a** and **15b** or **16b**, were also not identified (Scheme 4). Starting materials were recovered.

Table 2
Rearrangement of 3-(ferrocenylmethoxy)–2-sulfanyl-1*H*-pyrroles **2** into 2-(ferrocenylmethyl)–2-sulfanyl-1,2-dihydro-3*H*-pyrrol-3-ones **7**.

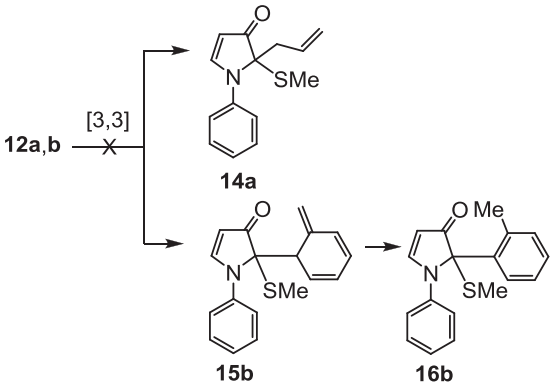


R1 = Me (**a**), *i*-Pr (**b**), CH_2=$CHOCH_2CH_2$ (**c**), *cyclo*-C_6H_9 (**d**), *cyclo*-C_7H_{13} (**e**), Ph (**f**), R2 = Me; R1 = CH_2=$CHOCH_2CH_2$, R2 = Et (**g**); R1 = Me, R2 = HC≡CCH_2 (**h**), Bn (**i**), $MeOC(=O)CH_2$ (**j**), *t*-$BuOC(=O)CH_2$ (**k**)

Pyrrole	Product of rearrangement		Time, days	Conversion of pyrrole **2**, %[a]
2a		**7a**	12 19	82 96
2b		**7b**	12	17
2c		**7c**	20 55	77 100
2d		**7d**	13 28	2 19
2e		**7e**	21	21
2f		**7f**	2 8 19	24 96 100
2g		**7g**	[b]	12
2h		**7h**	15	42
2i		**7i**	16	100
2j		**7j**	10 4.5[c]	16 90
2k		**7k**	4[c]	100
2l		**7l**	14	0

[a] Conversions of pyrroles **2** were calculated from ^{1}H NMR spectra. [b] During purification on neutral Al_2O_3. [c] The NMR spectra were recorded after 4.5 and 4 months, but the actual time required for 90 and 100% conversion of the corresponding pyrroles to 1,2-dihydro-3*H*-pyrrol-3-ones **7j,k** is not defined.

[a] Conversions of pyrroles **2** were calculated from ^{1}H NMR spectra.
[b] During purification on neutral Al_2O_3.
[c] The NMR spectra were recorded after 4.5 and 4 months, but the actual time required for 90 and 100% conversion of the corresponding pyrroles to 1,2-dihydro-3*H*-pyrrol-3-ones **7j,k** is not defined.



Scheme 4. Expected products of the Claisen rearrangement.

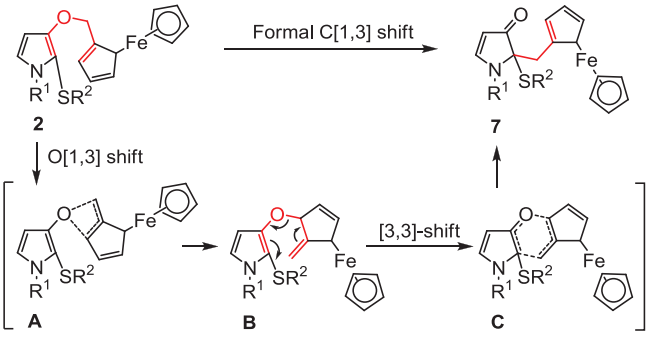
Thus, it becomes obvious that the ability of the pyrroles **2** to a rearrange into 1,2-dihydro-3*H*-pyrrol-3-ones **7**, moreover under such mild conditions, is provided by the specific effect of ferrocene fragment. According to the propensity to rearrange into 1,2-dihydro-3*H*-pyrrol-3-ones **7** under comparable conditions ($CDCl_3$, r.t., 19–21 days), pyrroles **2** stand in the following order (Table 2): **2f** (R^1 = Ph, 100%) > **2a** (R^1 = Me, ~96%) > **2c** (R^1 = CH_2=$CHOCH_2CH_2$, 77%) > **2e** (R^1 = cycloC_7H_{13}, 21%). Pyrrole **2i** (R^2 = Bn) quantitatively rearranges into 1,2-dihydro-3*H*-pyrrol-3-one **7i** after 16 days. While, pyrrole **2l** (R^2 = CH_2CN) did not show a tendency to rearrange into 1,2-dihydro-3*H*-pyrrol-3-one **7l**, when its solution in $CDCl_3$ was stored at room temperature for 14 days (analyzed by ^{1}H NMR). In the case of pyrrole **2 g**, on the contrary, the rearrangement into 1,2-dihydro-3*H*-pyrrol-3-one **7 g** occurs already when it is purified by column chromatography on neutral alumina (Table 2).

On the basis of the structure of the products obtained, which is confirmed by NMR spectra, it is possible to assume, that transformation of 3-(ferrocenylmethoxy)–1*H*-pyrroles **2** into 1,2-dihydro-3*H*-pyrrol-3-ones **7** occurs through the so-called «formal [1,3]-O-to-C rearrangement», a relatively rare type of rearrangements (predominantly in the range of allyl aryl ethers and hetarylmethyl vinyl ethers) [19]. The pyrrole ring is involved in this rearrangement as the vinyl ether component, whereas the ferrocenylmethyl substituent participates as the “allylic” counterpart. It should be emphasized that so far the rearrangements of this type among the pyrrole derivatives were not observed. It was also shown, that some other heteroaromatics, for instance, 1,3-oxazole [19h], furan, and thiophene [19i], participate as the vinyl ether component in the thermal (xylene, 120 °C, 3 h) [1,3]-O-to-C rearrangement of 5-(benzyloxy)–1,3-oxazoles and 2-(benzyloxy)furans and -thiophenes.

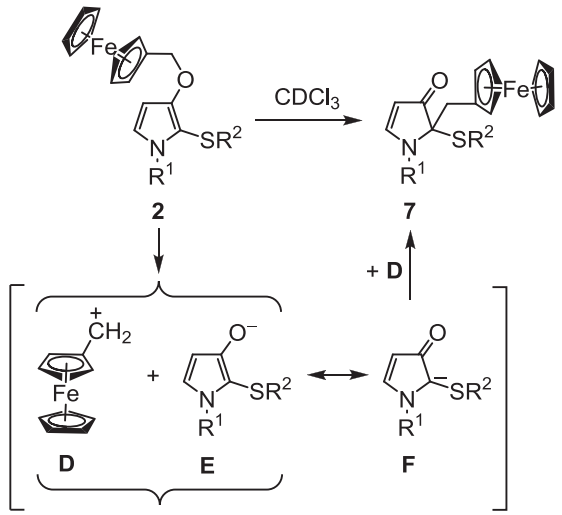
One of possible routes of the found rearrangement of pyrroles **2**, similarly to the published rearrangements of this type [19], includes a tandem O[1,3]-sigmatropic shift (through intermediate **A** with a configuration inversion of the oxygen atom) and [3,3]-sigmatropic shift (the Claisen rearrangement) in the intermediate **B** (via intermediate **C**) (Scheme 5).

However, most of the known examples of the formal [1,3]-O-to-C migration, occurring via the depicted in Scheme 5 plausible mechanism, belong to the thermally induced rearrangements [19a], whereas the rearrangement found by us is implemented at room temperature and is initiated, apparently, by HCl, presented in deuterium chloroform in trace amounts (see Table 2).

It can be also assumed that the transformation of pyrroles **2** into 1,2-dihydro-3*H*-pyrrol-3-ones **7** occurs through the catalytic or heterolytic carbon–oxygen bond cleavage [20], which leads to fairly stable ferrocenylmethyl carbocation **D** and pyrrolyloxide anions **E** (Scheme 6). The subsequent interaction between the carbocation **D**



Scheme 5. A plausible mechanism of rearrangement of pyrroles **2** into 1,2-dihydro-3*H*-pyrrol-3-ones **7** via a formal C[1,3] shift (involving tandem O[1,3]- and [3,3]-sigmatropic shifts).



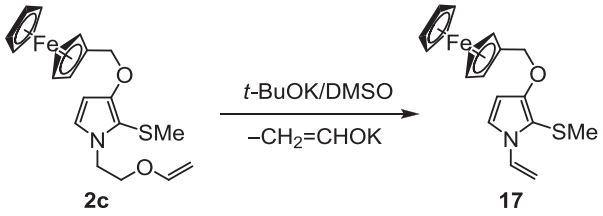
Scheme 6. Plausible mechanism of rearrangement of pyrroles **2** into 1,2-dihydro-3*H*-pyrrol-3-ones **7** in a $CDCl_3$ solution.

and the mesomeric pyrrolyl carbanion **F** gives rearrangement product **7**.

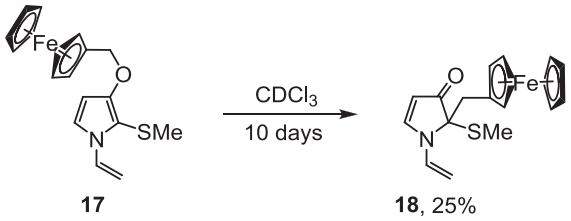
The presence in the mass spectra of pyrroles **2** of the peak of an ion with a mass of 199 of maximum intensity, $[FcCH_2]^+$ (100%), serves as an additional confirmation of the ease of cleavage of the C–O bond leading to the formation of the stable ferrocenylmethyl carbocation **D** (Scheme 6).

The proposed mechanism of the rearrangement of **2**→**7** in $CDCl_3$ (via ferrocenylmethyl carbocation) is also supported by the literature data [19d,19i,20a,20b]. Thus, [1,3]-benzyl group migration from oxygen to carbon in $CDCl_3$ (r.t., 6 – 8 weeks) was observed for 2-furylbenzyl ethers [19i]. 2-Furylmethyl and 2-thienylmethyl vinyl ethers in medium of lithium perchlorate in diethyl ether (r.t., 6 h) were smoothly transformed to the formal [1,3]-rearrangement products, γ-heteroaryl propanals [19d]. In contrast, benzyl and naphthylmethyl vinyl ethers exhibited the inertness under similar conditions. Formation of rather stable ferrocenylmethyl carbocations under very mild conditions (through heterolytic cleavage of the C–O bond in α-ferrocenylalkyl carbonates) and their participation in ferrocenylalkylation reactions was also described [20a,20b].

We have also tested the ability of pyrroles **2** to thermally-induced rearrangement into 1,2-dihydro-3H-pyrrol-3-ones **7** (Scheme 5) or **10**/**11** (Scheme 2). For this, a solution of pyrrole **2f** in toluene was heated at 98–100 °C for several hours. The course of the reaction was monitored by ^{1}H NMR spectra. It turned out that, after 3 h, the conversion of pyrrole **2f** into 3H-pyrrol-3-one **7f** was 45% and after 8 h – 96%. Products of



Scheme 7. Transformation of pyrrole **2c** into *N*-vinylpyrrole **17** via superbase-induced cleavage of C–O bond.



Scheme 8. Rearrangement of pyrrole **17** into 1,2-dihydro-3*H*-pyrrol-3-one **18**.

Claisen rearrangement, i.e., compounds **10**/**11**, were not identified. Thus, the experimental results obtained allow us to consider the transformation of pyrroles **2** into 1,2-dihydro-3H-pyrrole-3-ones **7** as the first example of implementation of the thermal formal [1,3]-O-to-C-sigmatropic rearrangement in a series of (ferrocenylmethoxy)pyrroles, occurring via the depicted in Scheme 5 plausible mechanism.

*2.3. Transformation of N-[2-(vinyloxy)ethyl]−3-(ferrocenylmethoxy)−1H-pyrrole **2c** into N-vinylpyrrole **17***

Recently, we have established that 3–methoxy-analogue of pyrrole **2c**, prepared from lithiated methoxyallene, 2-(vinyloxy)ethyl isothiocyanate and MeI, by treatment with

t-BuOK/DMSO can be easily converted into the 3–methoxy-2-(methylsulfanyl)-*N*-vinylpyrrole [10b]. The involvement of 3-(ferrocenylmethoxy)pyrrole **2c** in the reaction with

t-BuOK/DMSO also gave *N*-vinylpyrrole, namely 3-(ferrocenylmethoxy)−2-(methylsulfanyl)−1-vinyl-1*H*-pyrrole (**17**) in a yield of 96%. The reaction proceeds at r.t. for ~30 min via elimination of vinyl alcohol from the *N*-[2-(vinyloxy)ethyl] substituent and represents a novel approach to ferrocenyl-substituted *N*-vinylpyrroles (Scheme 7).

N-Vinylpyrrole **17**, like pyrroles **2**, in $CDCl_3$ solution rearranges into 1,2-dihydro-3*H*-pyrrol-3-one **18** (Scheme 8).

3. Conclusion

We have described a one- and two-pot assembly of 3-(ferrocenylmethoxy)−2-sulfanyl-1*H*-pyrroles from readily accessible starting materials, lithiated (ferrocenylmethoxy)allene, isothiocyanates, and alkylating agents. The reaction proceeds via synthesis and Cu(I)-catalyzed intramolecular cyclization of alkyl 2-(ferrocenylmethoxy)buta-2,3-dienimidothioates (1-aza-1,3,4-trienes), which serve as the key intermediates in this approach. Although the non-optimized isolated yields of the target products were from moderate to good (18–67%), the presented approach allows direct synthesis of 3-(ferrocenylmethoxy)−2-sulfanyl-1*H*-pyrroles as promising basic compounds for the design of new generations of molecular sensors, ferromagnets, catalysts, optoelectronic materials, drugs, and diagnostic tools for medicine. An unexpected rearrangement of the synthesized 3-(ferrocenylmethoxy)−2-sulfanyl-1*H*-pyrroles into 2-(ferrocenylmethyl)−2-sulfanyl-1,2-dihydro-3*H*-pyrrol-3-ones, occurring at room temperature in $CDCl_3$ or when heated in toluene, has been discovered. On the basis of experimental and literature data it is assumed that the rearrangement proceeds through two different mechanisms of activation of the formal [1,3]-O-to-C-migration of ferrocenylmethyl group: 1) via acid-mediated process with generation of ferrocenylmethyl carbocation (in $CDCl_3$) or 2) via tandem O[1,3]-sigmatropic shift and Claisen rearrangement (under heating). Although pyrroles **2** can be classified as Claisen systems, no [3,3]-sigmatropic process was observed, in all cases.

4. Experimental

4.1. General

(Ferrocenylmethoxy)allene (**1**) was synthesized using the method reported earlier [11]. Isopropyl, 2-(vinyloxy)ethyl, cyclopentyl, and cycloheptyl isothiocyanates were obtained according to our procedure from primary amines, carbon disulfide and acetyl chloride [9a,21a] or acetic anhydride [9a,21b]. (Allyloxy)- and (benzyloxy)allenes were prepared as described in literature [22]. Methyl and phenyl isothiocyanates, *n*-BuLi (1.6, 2.37, and 2.5 M solutions in hexane), and the other employed reagents and solvents are commercially available. THF was dried and purified by standing over dispersed KOH (~50 g/L), boiling, and distillation over Na in the presence of benzophenone under argon atmosphere. All other solvents were purified according to standard procedures. All reactions were performed under anhydrous conditions and in an argon atmosphere. For all reactions at low temperatures a cooling bath with liquid nitrogen was used. All reactions were monitored by TLC on precoated with Merck silica gel 60 F254 plates, were visualised by exposure to iodine vapor. Yields were referred to isolated products.

The ^{1}H (400.13 MHz), ^{13}C (100.62 MHz), and ^{15}N (40.55 MHz) NMR spectra were recorded on a Bruker DPX-400 and Bruker AV-400 spectrometers in $CDCl_3$ solution at an ambient temperature. Chemical shifts (δ) are quoted in ppm, to the nearest 0.01 ppm, and are referenced to HMDS (for ^{1}H and ^{13}C) and $MeNO_2$ (for ^{15}N) as internal standards. Coupling constants (J) are reported in Hertz (Hz) to the nearest 0.1 Hz. Assignments of spectra were carried out using 2D experiments. IR spectra were measured neat or as KBr pellets on a Bruker Vertex-70 infrared spectrophotometer. The mass spectra (electron impact, 70 eV, direct input of the sample to the ion source) were recorded on a Shimadzu GCMS-QP5050A instrument. The microanalyses were performed on a Flash EA 1112 Series elemental analyzer. Melting points were determined using a SGW X-4 Melting-point apparatus with microscope.

4.2. X-ray crystallography

X-ray crystallography study of the pyrrole **2f** was carried out on a Bruker Kappa Apex II CCD diffractometer using φ,ω-scans of narrow (0.5 °) frames with Mo$K\alpha$ radiation (λ = 0.71073 Å) and a graphite monochromator. The structure was solved by direct methods and refined by full-matrix least-squares method against all F2 in anisotropic approximation using the SHELX-97 programs set [23]. The hydrogen atom positions were calculated geometrically. Absorption corrections were applied using the empirical multiscan method with the SADABS program [24]. CCDC 1,015,374 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via http://www.ccdc.cam.ac.uk/cgi-bin/catreq.cgi, or from the Cambridge Crystallographic Data center, 12 Union Road, Cambridge CB2 1EZ, UK; fax: (+44) 1223 336 033; or e-mail: deposit@ccdc.cam.ac.uk.

4.3. General procedure: preparation of 3-(ferrocenylmethoxy)−2-sulfanyl-1H-pyrroles **2** (Table 1) and their rearrangement into 2-(ferrocenylmethyl)−2-sulfanyl-1,2-dihydro-3H-pyrrol-3-ones **7** (Table 2)

To a vigorously stirred solution of (ferrocenylmethoxy)allene (**1**) (10.0 – 15.0 mmol) in dry THF (20–40 mL), *n*-BuLi (6 – 11 mL of 1.6, 2.37, and 2.5 M solutions in hexane, 15.0 – 17.6 mmol) was added at −100 to −90 °C under argon. After stirring for an additional 5 – 15 min at ca. −65 to −40 °C, the solution was cooled to ca. −90 to −70 °C and a mixture of isothiocyanate (10.0 – 15.0 mmol) and THF (3 – 5 mL) was added in one portion. After being stirred for 5 – 10 min at −50 to −40 °C (for methyl isothiocyanate), for 20 min at ca. −30 °C (for iso and cycloalkyl isothiocyanates), for 10 min at −45 to −40 °C [for 2-(vinyloxy)ethyl isothiocyanate] and for 15 min at ca. −65 °C (for phenyl isothiocyanate), the mixture was again cooled to ca. −80 °C and methyl iodide (22.9 – 35.7 mmol, 1.5 – 2.4 equiv) or propargyl bromide (18.5 mmol, 1.2 equiv) or benzyl bromide (10.0 mmol, 1.0 equiv) were added in one portion. The cooling bath was removed, and when the temperature was reached −10 to −5 °C, CuI (2.10−3.15 mmol) was added. The reaction mixture was stirred for 15−50 min at 32−50 °C, then cooled to r.t. and treated with saturated aqueous NH_4Cl solution (30 mL) contained NaCN (0.50 g). The layers were separated, and the aqueous phase was extracted with Et_2O (3 × 30 mL). The organic layers were combined, washed with H_2O (50 mL), dried ($MgSO_4$), and concentrated under reduced pressure to give a residue consisting of corresponding pyrrole **2** (by ^{1}H NMR). Pure compounds **2a**–**2i** were isolated by column chromatography on neutral alumina, using hexane and hexane/Et_2O as eluent. Yields (based on isothiocyanate) and characteristic data of pyrroles **2a**–**2i** are reported below.

Compounds **2j, 2k**, and **2l** were obtained similarly, but in two preparative steps, namely, with the isolation of the corresponding 1-aza-1,3,4-triene **5** followed by its cyclization into pyrrole in the presence of CuBr. If the reaction was carried out in one preparative stage, i.e., the catalyst was added to the reaction mixture containing in situ generated 1-aza-1,3,4-triene **5j** or **5k**, then, instead of the expected pyrroles **2j** and **2k**, the corresponding thiophenes **6j** and **6k** were obtained [14].

During storage of a solutions of pyrroles **2a**–***k*** in $CDCl_3$ at room temperature their rearrangement (partial or complete) into 2-(ferrocenylmethyl)−2-sulfanyl-1,2-dihydro-3*H*-pyrrol-3-ones **7** (Table 2) was observed (by ^{1}H NMR spectra). The TLC analysis indicated that R_f values of products **2** and **7** are very close and difficult to separate by column chromatography. When the reaction was conducted in toluene at 98−100 °C for 8 h, pyrrole **2f** is almost quantitatively converted into 1,2-dihydro-3*H*-pyrrol-3-one **7f**.

4.3.1. 3-(Ferrocenylmethoxy)−1-methyl-2-(methylsulfanyl)−1H-pyrrole (***2a***)

The general procedure starting from allene **1** (3.78 g, 14.9 mmol)/THF (40 mL), *n*-BuLi (10 mL of 1.6 M solution in hexane, 16.0 mmol), MeNCS (1.09 g, 14.9 mmol)/THF (3 mL), MeI (3.25 g, 22.9 mmol, 1.5 equiv), and CuI (0.60 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1, 3:1, 1:1), 2.81 g (55%) of **2a** as yellow needles, m.p. 89−90 °C; [Found: C, 59.90; H, 5.53; N, 4.11; S, 9.35. $C_{17}H_{19}FeNOS$ (341.249) requires C, 59.83; H, 5.61; N, 4.10; S, 9.40%]. IR (KBr): 3124 (w), 3099 (w), 3086 (w), 3078 (w), 2983 (w), 2916 (m), 2860 (w), 2821 (w), 1634 (w), 1544 (s), 1535 (sh), 1488 (sh), 1480 (w), 1468 (w), 1456 (w), 1427 (sh), 1413 (m), 1393 (s), 1376 (w), 1352 (w), 1320 (s), 1242 (w), 1230 (w), 1115 (w), 1102 (m), 1073 (s), 1049 (m), 1028 (w), 1017 (w), 1002 (w), 966 (m), 957 (m), 927 (w), 880 (w), 867 (w), 837 (m), 816 (m), 798 (m), 728 (w), 720 (m), 700 (s), 660 (w), 644 (w), 627 (w), 604 (w), 529 (w), 505 (s), 485 (s), 420 (w) cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.52 (d, ^3J = 3.2 Hz, 1H, 5-H), 5.86 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.77 (s, 2H, OCH_2Fc), 4.28 (nr m, 2H, 2′,5′-H, Fc), 4.15 (s, 5H, C_5H_5, Fc), 4.12 (nr m, 2H, 3′,4′-H, Fc), 3.58 (s, 3H, NCH_3), 2.11 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 150.1 (3-C), 121.0 (5-C), 108.1 (2-C), 96.6 (4-C), 83.4 (1′-C, Fc), 69.9 (OCH_2Fc), 69.2 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-C, Fc), 34.1 (NCH_3), 20.4 (SCH_3) ppm. The ^{1}H,^{13}C HMBC 2D experiment provided additional support for the proposed structure. ^{1}H,^{15}N HMBC: δ −249.3 ppm. MS-EI (70 eV), *m/z* (%): 341 $[M]^{+\bullet}$ (2), 199 $[FcCH_2]^+$ (100), 121 $[C_5H_5Fe]^+$ (40), 56 $[Fe]^+$ (22).

4.3.1.1. 2-(Ferrocenylmethyl)−1-methyl-2-(methylsulfanyl)−1,2-dihydro-3H-pyrrol-3-one (***7a***) (Tentative assignment) (Table 2).

^{1}H NMR (400 MHz, $CDCl_3$): δ 7.57 (br s, 1H, 5-H), 4.95 (br s, 1H, 4-H), 4.16, 4.12 (br s, 2H, 2′,5′-H, Fc), 4.06 (s, 5H, C_5H_5, Fc), 4.01, 3.97 (br s, 2H, 3′,4′-H, Fc), 3.11, 2.75 (both d, ^2J = 16.0 Hz, 2H, CH_2Fc), 3.06 (s, 3H, NCH_3), 1.75 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 200.4 (C=O), 165.1 (5-C), 97.5 (4-C), 79.9 (1′-C, Fc), 76.3 (2-C), 70.0, 68.8 (2′,5′-C, Fc), 68.7 (C_5H_5, Fc), 68.1, 67.4 (3′,4′-C, Fc), 35.5 (CH_2Fc), 32.6 (NCH_3), 11.4 (SCH_3) ppm.

4.3.2. 3-(Ferrocenylmethoxy)−1-isopropyl-3-2-(methylsulfanyl)−1H-pyrrole (***2b***)

The general procedure starting from allene **1** (3.78 g, 14.9 mmol)/THF (40 mL), n-BuLi (10 mL of 1.6 M solution in hexane, 16.0 mmol), i-PrNCS (1.51 g, 14.9 mmol)/THF (5 mL), MeI (3.75 g, 26.4 mmol, 1.8 equiv), and CuI (0.60 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1, 3:1, 1:1), 2.41 g (44%) of **2b** as orange crystals, m.p. 70−71 °C; [Found: C, 61.69; H, 6.23; N, 3.71; S, 8.57. $C_{19}H_{23}FeNOS$ (369.303) requires C, 61.79; H, 6.28; N, 3.79; S, 8.68%]. IR (KBr): 3095 (sh), 3084 (w), 2979 (w), 2920 (w), 2909 (sh), 2857 (w), 2827 (w), 1635 (w), 1585 (w), 1540 (s), 1492 (w), 1480 (w), 1453 (m), 1415 (m), 1393 (m), 1383 (sh), 1368 (m), 1316 (s), 1268 (m), 1240 (w), 1215 (w), 1128 (w), 1106 (m), 1082 (s), 1062 (s), 1039 (w), 1026 (sh), 1019 (w), 999 (m), 964 (m), 949 (sh), 931 (w), 903 (w), 880 (w), 860 (w), 846 (w), 820 (m), 809 (m), 798 (m), 698 (s), 656 (w), 643 (w), 624 (w), 578 (w), 510 (m), 481 (s), 443 (w), 421 (w) cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.62 (d, ^3J = 3.2 Hz, 1H, 5-H), 5.92 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.76 (s, 2H, OCH_2Fc), 4.75−4.68 [m, 1H, NCH$(CH_3)_2$], 4.29 (t, ^3J = 1.8 Hz, 2H, 2′,5′-H, Fc), 4.16 (s, 5H, C_5H_5, Fc), 4.12 (t, ^3J = 1.8 Hz, 2H, 3′,4′-H, Fc), 2.13 (s, 3H, SCH_3), 1.36 [d, ^3J = 6.7 Hz, 6H, NCH$(CH_3)_2$] ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 150.0 (3-C), 115.9 (5-C), 106.8 (2-C), 96.5 (4-C), 83.6 (1′-C, Fc), 69.6 (OCH_2Fc), 69.1 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-C, Fc), 46.7 [NCH$(CH_3)_2$], 23.6 [NCH$(CH_3)_2$], 20.7 (SCH_3) ppm. MS-EI (70 eV), *m/z* (%): 369 $[M]^{+\bullet}$ (2), 199 $[FcCH_2]^+$ (100), 121 $[C_5H_5Fe]^+$ (49), 56 $[Fe]^+$ (20).

4.3.2.1. 2-(Ferrocenylmethyl)−1-isopropyl-2-(methylsulfanyl)−1,2-dihydro-3H-pyrrol-3-one (***7b***) (Tentative assignment) (Table 2).

^{1}H NMR (400 MHz, $CDCl_3$): δ 7.79 (br d, ^3J = 2.5 Hz, 1H, 5-H), 5.03 (br d, ^3J = 2.5 Hz, 1H, 4-H), 4.71 [sept, ^3J = 6.6 Hz, 1H, NCH$(CH_3)_2$], 4.15 (br s, 2H, 2′,5′-H, Fc), 4.04 (s, 5H, C_5H_5, Fc), 3.97 (br s, 2H, 3′,4′-H, Fc), 3.19, 2.80 (both d, ^2J = 14.3 Hz, 2H, CH_2Fc), 1.80 (s, 3H, SCH_3), 1.28, 1.14 [both d, ^3J = 6.6 Hz, 6H, NCH$(CH_3)_2$] ppm.

*4.3.3. 3-(Ferrocenylmethoxy)–2-(methylsulfanyl)–1-[2-(vinyloxy)ethyl]–1H-pyrrole (**2c**)*

The general procedure starting from allene **1** (3.81 g, 15.0 mmol)/THF (25 mL), n-BuLi (6 mL of 2.5 M solution in hexane, 15.0 mmol), CH_2=$CHOCH_2CH_2NCS$ (1.94 g, 15.0 mmol)/THF (5 mL), MeI (4.20 g, 29.6 mmol, 2.0 equiv), and CuI (0.60 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1), 3.97 g (67%) of **2c** as yellow needles, m.p. 79–80 °C; [Found: C, 60.29; H, 5.75; N, 3.45; S, 7.98. $C_{20}H_{23}FeNO_2S$ (397.312) requires C, 60.46; H, 5.83; N, 3.53; S, 8.07%]. IR (KBr): 3114 (w), 2972 (sh), 2921 (w), 2873 (w), 1623 (s), 1539 (s), 1473 (w), 1434 (sh), 1395 (s), 1322 (s), 1237 (sh), 1206 (s), 1099 (sh), 1063 (s), 1030 (m), 1008 (sh), 957 (s), 823 (s), 760 (w), 730 (m), 659 (w), 611 (w), 492 (s), 425 (w) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 6.65 (d, ^3J = 3.2 Hz, 1H, 5-H), 6.38 (dd, $^3J_{trans}$ = 14.2 Hz, $^3J_{cis}$ = 6.7 Hz, 1H, OCH=CH_2), 5.90 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.77 (s, 2H, OCH_2Fc), 4.28 (t, ^3J = 1.8 Hz, 2H, 2′,5′-H, Fc), 4.20 (t, ^3J = 5.6 Hz, 2H, NCH_2CH_2O), 4.16 (s, 5H, C_5H_5, Fc), 4.16 (dd, $^3J_{trans}$ = 14.2 Hz, $^2J_{gem}$ = 2.2 Hz, 1H, OCH=CH_2), 4.12 (t, ^3J = 1.8 Hz, 2H, 3′,4′-H, Fc), 3.99 (dd, $^3J_{cis}$ = 6.7 Hz, $^2J_{gem}$ = 2.2 Hz, 1H, OCH=CH_2), 3.87 (t, ^3J = 5.6 Hz, 2H, NCH_2CH_2O), 2.12 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 151.3 (OCH=CH_2), 150.4 (3-C), 121.1 (5-C), 107.3 (2-C), 97.0 (4-C), 87.1 (OCH=CH_2), 83.4 (1′-C, Fc), 69.7 (OCH_2Fc), 69.2 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-H, Fc), 67.7 (NCH_2CH_2O), 45.9 (NCH_2CH_2O), 20.6 (SCH_3) ppm. ^1H,^{15}N HMBC: δ −228.8 ppm. MS-EI (70 eV), *m/z* (%): 397 $[M]^{+\bullet}$ (2), 199 $[FcCH_2]^+$ (100), 121 $[C_5H_5Fe]^+$ (48), 56 $[Fe]^+$ (29).

*4.3.3.1. 2-(Ferrocenylmethyl)–2-(methylsulfanyl)–1-[2-(vinyloxy)ethyl]–1,2-dihydro-3H-pyrrol-3-one (**7c**) (Tentative assignment) (Table 2).* ^1H NMR (400 MHz, $CDCl_3$): δ 7.80 (br s, 1H, 5-H), 6.43 (dd, $^3J_{trans}$ = 14.2 Hz, $^3J_{cis}$ = 6.6 Hz, 1H, O*CH*=CH_2), 5.00 (br s, 1H, 4-H), 4.26–3.93 (m, 10H, OCH=*CH*$_2$, N*CH*$_2$*CH*$_2$O, 2′,3′,4′,5′-H, Fc), 4.06 (s, 5H, C_5H_5, Fc), 3.12, 2.75 (both d, ^2J = 14.4 Hz, 2H, *CH*$_2$Fc), 1.78 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.8 (C=O), 165.4 (5-C), 150.9 (OCH=CH_2), 98.1 (4-C), 87.7 (OCH=*CH*$_2$), 79.8 (1′-C, Fc), 70.0 (2-C), 69.0, 68.7 (2′,5′-C, Fc), 68.8 (C_5H_5, Fc), 68.1, 67.5 (3′,4′-C, Fc), 66.5 (NCH_2CH_2O), 44.4 (N*CH*$_2$CH$_2$O), 36.3 (*CH*$_2$Fc), 11.4 (SCH_3) ppm.

*4.3.4. 1-Cyclopentyl-3-(ferrocenylmethoxy)–2-(methylsulfanyl)–1H-pyrrole (**2d**)*

The general procedure starting from allene **1** (3.78 g, 14.9 mmol)/THF (40 mL), *n*-BuLi (11 mL of 1.6 M solution in hexane, 17.6 mmol), cycloC_5H_9NCS (1.89 g, 14.9 mmol)/THF (5 mL), MeI (4.07 g, 28.7 mmol, 1.9 equiv), and CuI (0.60 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1), 1.31 g (22%) of **2d** as yellow crystals, m.p. 58–60 °C; [Found: C, 63.69; H, 6.23; N, 3.65; S, 8.17. $C_{21}H_{25}FeNOS$ (395.340) requires C, 63.80; H, 6.37; N, 3.54; S, 8.11%]. IR (KBr): 3094 (s), 2959 (s), 2918 (s), 2871 (s), 1633 (w), 1545 (s), 1528 (sh), 1475 (w), 1452 (w), 1412 (m), 1397 (s), 1380 (sh), 1333 (s), 1307 (s), 1264 (w), 1238 (m), 1204 (w), 1165 (w), 1115 (sh), 1106 (s), 1075 (vs), 1039 (m), 1024 (m), 1001 (m), 967 (s), 924 (m), 900 (sh), 885 (w), 857 (sh), 820 (vs), 749 (w), 707 (s), 663 (w), 618 (w), 501 (sh), 484 (vs) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 6.62 (d, ^3J = 3.2 Hz, 1H, 5-H), 5.90 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.88–4.81 (m, 1H, NCH), 4.76 (s, 2H, O*CH*$_2$Fc), 4.29 (t, ^3J = 1.8 Hz, 2H, 2′,5′-H, Fc), 4.16 (s, 5H, C_5H_5, Fc), 4.12 (t, ^3J = 1.8 Hz, 2H, 3′,4′-H, Fc), 2.13 (s, 3H, SCH_3), 2.13–2.06, 1.82–1.71 (m, 4H, 2′′,5′′-H), 1.69–1.66 (m, 4H, 3′′,4′′-H) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 150.1 (3-C), 116.4 (5-C), 107.8 (2-C), 96.6 (4-C), 83.6 (1′-C, Fc), 69.7 (OCH_2Fc), 69.1 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-C, Fc), 56.6 (NCH), 33.6 (2′′,5′′-C), 24.2 (3′′,4′′-C), 20.7 (SCH_3) ppm.

*4.3.4.1. 1-Cyclopentyl-2-(ferrocenylmethyl)–2-(methylsulfanyl)–1,2-dihydro-3H-pyrrol-3-one (**7d**) (Tentative assignment) (Table 2).* ^1H NMR (400 MHz, $CDCl_3$): δ 7.76 (br s, 1H, 5-H), 5.00 (br s, 1H, 4-H), 4.83 (nr m, 1H, NCH), 4.16 (nr m, 2H, 2′,5′-H, Fc), 4.11 (s, 5H, C_5H_5, Fc), 3.96 (nr m, 2H, 3′,4′-H, Fc), 3.14, 2.78 (both d, ^2J = 14.3 Hz, 2H, *CH*$_2$Fc), 2.09 (nr m, 4H, 2′′,5′′-H), 1.79 (s, 3H, SCH_3), 1.68 (nr m, 4H, 3′′,4′′-H) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.7 (C=O), 161.6 (5-C), 97.7 (4-C), 80.1 (1′-C, Fc), 70.0, 68.4 (2′,5′-C, Fc), 69.3 (2-C), 68.7 (C_5H_5, Fc), 68.0, 67.1 (3′,4′-C, Fc), 56.1 (NCH), 36.5 (*CH*$_2$Fc), 34.9, 32.5 (2′′,5′′-C), 24.1, 23.8 (3′′,4′′-C), 11.7 (SCH_3) ppm.

*4.3.5. 1-Cycloheptyl-3-(ferrocenylmethoxy)–2-(methylsulfanyl)–1H-pyrrole (**2e**)*

The general procedure starting from allene **1** (3.78 g, 14.9 mmol)/THF (40 mL), *n*-BuLi (11 mL of 1.6 M solution in hexane, 17.6 mmol), cyclo$C_7H_{13}NCS$ (2.31 g, 14.9 mmol)/THF (5 mL), MeI (4.37 g, 30.8 mmol, 2.1 equiv), and CuI (0.60 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1), 2.32 g (37%) of **2e** as yellow-orange prisms, m.p. 63.5 – 65 °C; [Found: C, 65.19; H, 6.73; N, 3.25; S, 7.47. $C_{23}H_{29}FeNOS$ (423.393) requires C, 65.25; H, 6.90; N, 3.31; S, 7.57%]. IR (KBr): 3094 (w), 2992 (w), 2923 (m), 2851 (m), 1546 (s), 1521 (w), 1508 (w), 1481 (w), 1458 (m), 1446 (w), 1413 (m), 1395 (s), 1380 (sh), 1362 (sh), 1352 (m), 1313 (s), 1284 (w), 1240 (w), 1223 (w), 1207 (w), 1173 (w), 1117 (w), 1105 (s), 1074 (s), 1053 (sh), 1040 (m), 1025 (m), 1014 (w), 1000 (m), 973 (m), 967 (m), 930 (w), 902 (w), 887 (w), 863 (w), 834 (s), 809 (s), 799 (s), 754 (w), 704 (s), 669 (sh), 661 (w), 620 (m), 593 (w), 574 (w), 496 (s), 483 (s), 430 (w), 397 (w) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 6.62 (d, ^3J = 3.2 Hz, 1H, 5-H), 5.88 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.75 (s, 2H, O*CH*$_2$Fc), 4.48 (tt, ^3J = 10.3 Hz, ^3J = 4.0 Hz, 1H, NCH), 4.29 (t, ^3J = 1.8 Hz, 2H, 2′,5′-H, Fc), 4.15 (s, 5H, C_5H_5, Fc), 4.12 (t, ^3J = 1.8 Hz, 2H, 3′,4′-H, Fc), 2.12 (s, 3H, SCH_3), 2.00–1.90 (m, 2H, 2′′$_{eq}$,7′′$_{eq}$-H), 1.82–1.50 (m, 10H, 2′′$_{ax}$,7′′$_{ax}$-H, 3′′–6′′-H) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 149.8 (3-C), 116.8 (5-C), 106.6 (2-C), 96.5 (4-C), 83.6 (1′-C, Fc), 69.7 (OCH_2Fc), 69.1 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-C, Fc), 56.7 (NCH), 36.4 (2′′,7′′-C), 27.7 (3′′,6′′-C), 24.9 (4′′,5′′-C), 20.8 (SCH_3) ppm. MS-EI (70 eV), *m/z* (%): 423 $[M]^{+\bullet}$ (2), 199 $[FcCH_2]^+$ (100), 121 $[C_5H_5Fe]^+$ (36), 56 $[Fe]^+$ (11).

*4.3.5.1. 1-Cycloheptyl-2-(ferrocenylmethyl)–2-(methylsulfanyl)–1,2-dihydro-3H-pyrrol-3-one (**7e**) (Tentative assignment) (Table 2).* ^1H NMR (400 MHz, $CDCl_3$): δ 7.78 (br s, 1H, 5-H), 5.03 (br s, 1H, 4-H), 4.14 (br s, 2H, 2′,5′-H, Fc), 4.09 (s, 5H, C_5H_5, Fc), 4.06 (br s, 2H, 3′,4′-H, Fc), 3.33–3.30 (m, 1H, NCH), 3.18, 2.82 (both d, ^2J = 14.9 Hz, 2H, *CH*$_2$Fc), 1.82 (s, 3H, SCH_3), 1.82–1.23 (m, 12H, C_7H_{12}) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.2 (C=O), 162.0 (5-C), 96.8 (4-C), 80.4 (1′-C, Fc), 70.1 (2-C), 69.4, 69.1 (2′,5′-C, Fc), 68.8 (C_5H_5, Fc), 68.1, 67.0 (3′,4′-C, Fc), 56.6 (NCH), 37.8 (7′′-C), 36.4 (*CH*$_2$Fc), 35.1 (2′′-C), 27.5 (6′′-C), 27.0 (3′′-C), 25.0 (5′′-C), 24.7 (4′′-C), 12.0 (SCH_3) ppm.

*4.3.6. 3-(Ferrocenylmethoxy)–2-(methylsulfanyl)–1-phenyl-1H-pyrrole (**2f**)*

The general procedure starting from allene **1** (3.78 g, 14.9 mmol)/THF (40 mL), *n*-BuLi (11 mL of 1.6 M solution in hexane, 17.6 mmol), PhNCS (2.00 g, 14.8 mmol)/THF (5 mL), MeI (5.07 g, 35.7 mmol, 2.4 equiv), and CuI (0.60 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1), 2.46 g (41%) of **2f** as yellow

plates, m.p. 110–112 °C; [Found: C, 65.40; H, 5.21; N, 3.39; S, 7.87. $C_{22}H_{21}FeNOS$ (403.319) requires C, 65.52; H, 5.25; N, 3.47; S, 7.95%]. IR (KBr): 3122 (w), 3097 (w), 3070 (w), 3041 (w), 2922 (w), 2868 (w), 2856 (w), 2826 (w), 1597 (m), 1541 (s), 1502 (s), 1480 (w), 1469 (w), 1455 (w), 1396 (s), 1367 (m), 1350 (s), 1318 (w), 1309 (w), 1254 (w), 1243 (w), 1231 (w), 1174 (w), 1161 (w), 1131 (w), 1106 (w), 1088 (w), 1072 (s), 1039 (w), 1029 (w), 1003 (w), 992 (m), 964 (w), 935 (w), 913 (w), 896 (w), 855 (w), 816 (m), 797 (w), 763 (s), 727 (s), 712 (w), 696 (s), 656 (w), 601 (w), 579 (w), 500 (w), 490 (w), 480 (m), 455 (w), 429 (w) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 7.44–7.40 (m, 2H, 3″,5″-H, NC_6H_5), 7.36–7.31 (m, 3H, 2″,4″,6″-H, NC_6H_5), 6.76 (d, ^3J = 3.2 Hz, 1H, 5-H), 6.06 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.85 (s, 2H, OCH_2Fc), 4.33 (t, ^3J = 1.7 Hz, 2H, 2′,5′-H, Fc), 4.18 (s, 5H, C_5H_5, Fc), 4.15 (t, ^3J = 1.7 Hz, 2H, 3′,4′-H, Fc), 1.98 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 151.3 (3-C), 139.9 (1″-C, NC_6H_5), 128.7 (3″,5″-C, NC_6H_5), 127.0 (4″-C, NC_6H_5), 126.2 (2″,6″-C, NC_6H_5), 121.6 (5-C), 108.8 (2-C), 98.4 (4-C), 83.3 (1′-C, Fc), 69.9 (OCH_2Fc), 69.2 (2′,5′-C, Fc), 68.6 (C_5H_5, Fc), 68.4 (3′,4′-C, Fc), 20.3 (SCH_3) ppm. ^1H,^{15}N HMBC: δ −211.8 ppm. MS-EI (70 eV), *m/z* (%): 403 $[M]^{+\bullet}$ (2), 199 $[FcCH_2]^+$ (100), 121 $[C_5H_5Fe]^+$ (20), 56 $[Fe]^+$ (17).

*4.3.6.1. 2-(Ferrocenylmethyl)–2-(methylsulfanyl)–1-phenyl-1,2-dihydro-3H-pyrrol-3-one (**7f**).* (a) A solution of pyrrole **2f** (0.11 g, 0.27 mmol) in $CDCl_3$ (1 mL) was kept at r.t. for 8 days. Then $CDCl_3$ was evaporated, the residue was dissolved in Et_2O, ethereal solution was passed through neutral alumina layer (1 cm). Removal of Et_2O, gave 0.05 g (45%) of **7f** as brown fine crystalline powder, m.p. >160 °C (decomposition). b) A solution of pyrrole **2f** (0.21 g, 0.52 mmol) in toluene (2 mL) was warmed at 98–100 °C for 8 h. Then Et_2O was added and a solution was passed through neutral alumina layer (1 cm). Removal of solvents, gave 0.18 g of **7f** as light-brown solid, containing 8% of pyrrole **2f**, in yield of 79%.

IR (KBr): 3128 (w), 3091 (w), 3061 (w), 3032 (w), 2985 (w), 2949 (w), 2918 (w), 1673 (vs) (C=O), 1637 (sh), 1597 (w), 1585 (sh), 1529 (vs), 1497 (s), 1458 (w), 1424 (m), 1402 (m), 1342 (w), 1311 (s), 1313 (s), 1298 (sh), 1255 (w), 1223 (s), 1162 (w), 1106 (s), 1081 (sh), 1036 (w), 1021 (w), 1001 (m), 983 (sh), 961 (w), 936 (w), 895 (w), 856 (w), 821 (m), 799 (w), 752 (s), 726 (sh), 692 (m), 650 (w), 628 (w), 594 (w), 563 (w), 499 (s), 484 (s) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 8.10 (d, ^3J = 3.6 Hz, 1H, 5-H), 7.43–7.35 (m, 4H, 2″,3″,5″,6″-H, NC_6H_5), 7.21–7.17 (m, 1H, 4″-H, NC_6H_5), 5.33 (d, ^3J = 3.6 Hz, 1H, 4-H), 4.05, 3.90 (both br s, 2H, 3′,4′-H, Fc), 3.97 (s, 5H, C_5H_5, Fc), 3.78, 3.22 (both br s, 2H, 2′,5′-H, Fc), 3.24, 3.05 (both d, ^2J = 14.4 Hz, 2H, CH_2Fc), 1.93 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 201.4 (C=O), 160.5 (5-C), 140.0 (1″-C, NC_6H_5), 129.5 (3″,5″-C, NC_6H_5), 124.8 (4″-C, NC_6H_5), 119.7 (2″,6″-C, NC_6H_5), 102.6 (4-C), 79.2 (1′-C, Fc), 70.3 (2-C), 69.8, 68.1 (3′,4′-C, Fc), 69.7, 67.1 (2′,5′-C, Fc), 68.7 (C_5H_5, Fc), 35.5 (CH_2Fc), 11.8 (SCH_3) ppm. ^1H,^{15}N HMBC: δ −251.4 ppm. MS-EI (70 eV), *m/z* (%): 403 $[M]^{+\bullet}$ (31), 310 (11), 245 (13), 199 $[FcCH_2]^+$ (100), 178 (26), 121 $[C_5H_5Fe]^+$ (47), 77 (34), 56 $[Fe]^+$ (42).

*4.3.7. 2-(Ethylsulfanyl)–3-(ferrocenylmethoxy)–1-[2-(vinyloxy)ethyl]–1H-pyrrole (**2 g**)*

The general procedure starting from allene **1** (3.81 g, 15.0 mmol)/THF (25 mL), n-BuLi (6 mL of 2.5 M solution in hexane, 15.0 mmol), CH_2=$CHOCH_2CH_2NCS$ (1.94 g, 15.0 mmol)/THF (5 mL), EtI (4.79 g, 30.7 mmol, 2.0 equiv), and CuBr (0.60 g) afforded, after workup and chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1), 2.80 g (45%) of **2 g** as dark-brown viscous liquid (contaminated with ~6% of 3H-pyrrol-3-one **7 g**). ^1H NMR (400 MHz, $CDCl_3$): δ 6.68 (d, ^3J = 3.2 Hz, 1H, 5-H), 6.38 (dd, $^3J_{trans}$ = 14.2 Hz, $^3J_{cis}$ = 6.7 Hz, 1H, OCH=CH_2), 5.91 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.76 (s, 2H, OCH_2Fc), 4.28 (t, ^3J = 1.8 Hz, 2H, 2′,5′-H, Fc), 4.21 (t, ^3J = 5.7 Hz, 2H, NCH_2CH_2O), 4.16 (s, 5H, C_5H_5, Fc), 4.16 (dd, $^3J_{trans}$ = 14.2 Hz, $^2J_{gem}$ = 2.3 Hz, 1H, OCH=CH_2), 4.12 (t, ^3J = 1.8 Hz, 2H, 3′,4′-H, Fc), 3.99 (dd, $^3J_{cis}$ = 6.7 Hz, $^2J_{gem}$ = 2.3 Hz, 1H, OCH=CH_2), 3.86 (t, ^3J = 5.7 Hz, 2H, NCH_2CH_2O), 2.52 (q, ^3J = 7.4 Hz, 2H, SCH_2CH_3), 1.12 (t, ^3J = 7.4 Hz, 3H, SCH_2CH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 151.3 (OCH=CH_2), 151.2 (3-C), 121.2 (5-C), 105.4 (2-C), 96.8 (4-C), 87.1 (OCH=CH_2), 83.5 (1′-C, Fc), 69.5 (OCH_2Fc), 69.1 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-H, Fc), 67.6 (NCH_2CH_2O), 45.8 (NCH_2CH_2O), 31.2 (SCH_2CH_3), 14.7 (SCH_2CH_3) ppm. ^1H,^{15}N HMBC: δ −228.1 ppm. MS-EI (70 eV), *m/z* (%): 411 $[M]^{+\bullet}$ (5), 199 $[FcCH_2]^+$ (100), 121 $[C_5H_5Fe]^+$ (18), 56 $[Fe]^+$ (5).

*4.3.7.1. 2-(Ethylsulfanyl)–2-(ferrocenylmethyl)–1-[2-(vinyloxy)ethyl]–1,2-dihydro-3H-pyrrol-3-one (**7 g**) (Tentative assignment) (Table 2).* ^1H NMR (400 MHz, $CDCl_3$): δ 7.77 (d, ^3J = 3.4 Hz, 1H, 5-H), 6.43 (dd, $^3J_{trans}$ = 14.2 Hz, $^3J_{cis}$ = 6.7 Hz, 1H, OC*H*=CH_2), 5.00 (d, ^3J = 3.4 Hz, 1H, 4-H), 4.06 (s, 5H, C_5H_5, Fc), 3.12, 2.75 (both d, ^2J = 14.2 Hz, 2H, CH_2Fc), 2.40, 2.13 (both dq, ^2J = 11.7 Hz, ^3J = 7.5 Hz, 2H, SCH_2CH_3), 1.10 (t, ^3J = 7.5 Hz, 3H, SCH_2CH_3) ppm. Signals of protons of OCH=CH_2, NCH_2CH_2O, and 2′,3′,4′,5′-H (Fc) groups are masked by intense signals of pyrrole **2 g**. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.0 (*C* = *O*), 165.1 (5-C), 150.9 (OCH=CH_2), 97.9 (4-C), 87.6 (OCH=CH_2), 79.7 (1′-C, Fc), 75.4 (2-C), 70.0, 68.70 (2′,5′-C, Fc), 68.74 (C_5H_5, Fc), 68.1, 67.5 (3′,4′-C, Fc), 66.6 (NCH_2CH_2O), 44.5 (NCH_2CH_2O), 36.7 (CH_2Fc), 22.7 (SCH_2CH_3), 13.4 (SCH_2CH_3) ppm. ^1H,^{15}N HMBC: δ −264.1 ppm.

*4.3.8. 3-(Ferrocenylmethoxy)–1-methyl-2-(prop–2-ynylsulfanyl)–1H-pyrrole (**2 h**)*

The general procedure starting from allene **1** (3.78 g, 14.9 mmol)/THF (40 mL), *n*-BuLi (6 mL of 2.5 M solution in hexane, 15.0 mmol), MeNCS (1.09 g, 14.9 mmol)/THF (3 mL), HC≡CCH_2Br (2.20 g, 18.5 mmol), and CuI (0.60 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1), 0.953 g (18%) of **2 h** as amber viscous liquid, n_D^{20} 1.6402; [Found: C, 62.40; H, 5.18; N, 3.63; S, 8.67. $C_{19}H_{19}FeNOS$ (365.271) requires C, 62.48; H, 5.24; N, 3.83; S, 8.78%]. IR (neat): 3289 (s) (≡CH), 3094 (s), 2921 (s), 2868 (m), 2116 (w) (C≡C), 1541 (s), 1494 (w), 1481 (w), 1462 (w), 1443 (w), 1419 (m), 1391 (s), 1375 (sh), 1353 (w), 1320 (s), 1268 (w), 1239 (s), 1170 (w), 1119 (w), 1105 (m), 1072 (s), 1040 (m), 1025 (w), 1001 (m), 962 (m), 923 (m), 862 (sh), 821 (s), 750 (sh), 714 (s), 681 (sh), 642 (s), 601 (w), 497 (sh), 484 (s), 463 (w) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 6.56 (d, ^3J = 3.1 Hz, 1H, 5-H), 5.86 (d, ^3J = 3.1 Hz, 1H, 4-H), 4.77 (s, 2H, OCH_2Fc), 4.27 (nr m, 2H, 2′,5′-H, Fc), 4.15 (s, 5H, C_5H_5, Fc), 4.12 (nr m, 2H, 3′,4′-H, Fc), 3.63 (s, 3H, NCH_3), 3.17 (d, ^4J = 2.6 Hz, 2H, SCH_2), 2.13 (t, ^4J = 2.6 Hz, 1H, ≡CH) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 151.4 (3-C), 122.3 (5-C), 105.2 (2-C), 96.1 (4-C), 83.2 (1′-C, Fc), 80.6 (C≡), 71.5 (≡CH), 69.7 (OCH_2Fc), 69.1 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-C, Fc), 34.5 (NCH_3), 25.3 (SCH_2) ppm. ^1H,^{15}N HMBC: δ −231.6 ppm. MS-EI (70 eV), *m/z* (%): 365 $[M]^{+\bullet}$ (6), 199 $[FcCH_2]^+$ (100), 121 $[C_5H_5Fe]^+$ (30), 56 $[Fe]^+$ (23).

*4.3.8.1. 2-(Ferrocenylmethyl)–1-methyl-2-(prop–2-ynylsulfanyl)–1,2-dihydro-3H-pyrrol-3-one (**7 h**) (Tentative assignment) (Table 2).* ^1H NMR (400 MHz, $CDCl_3$): δ 7.53 (br s, 1H, 5-H), 4.98 (br s, 1H, 4-H), 4.16, 4.12 (br s, 2H, 2′,5′-H, Fc), 4.06 (s, 5H, C_5H_5, Fc), 4.01, 3.98 (br s, 2H, 3′,4′-H, Fc), 3.64 (s, 3H, NCH_3), 3.17 (d, ^4J = 2.5 Hz, 2H, SCH_2), 3.00, 2.76 (both d, ^2J = 14.2 Hz, 2H, CH_2Fc), 2.11 (t, ^4J = 2.5 Hz, 1H, ≡CH) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.2 (*C* = *O*), 165.9 (5-C), 97.3 (4-C), 83.2 (C≡), 79.2 (1′-C, Fc), 76.1

(2-C), 70.6 (≡CH), 69.1, 68.8 (2′,5′-C, Fc), 68.7 (C_5H_5, Fc), 68.2, 67.4 (3′,4′-C, Fc), 35.5 (OCH_2Fc), 33.1 (NCH_3), 16.9 (SCH_2) ppm.

*4.3.9. 2-(Benzylsulfanyl)−3-(ferrocenylmethoxy)−1-methyl-1H-pyrrole (**2i**)*

The general procedure starting from allene **1** (2.54 g, 10.0 mmol)/THF (20 mL), *n*-BuLi (4.5 mL of 2.5 M solution in hexane, 11.2 mmol), MeNCS (0.73 g, 10.0 mmol)/THF (3 mL), $PhCH_2Br$ (1.71 g, 10.0 mmol, 1.0 equiv), and CuI (0.40 g) afforded, after workup, passing of ethereal solution through neutral alumina layer (1.5 cm), and further chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1), 2.07 g (50%) of **2i** as brown viscous liquid; [Found: C, 66.33; H, 5.42; N, 3.44; S, 7.55. $C_{23}H_{23}FeNOS$ (417.345) requires C, 66.19; H, 5.56; N, 3.36; S, 7.68%]. IR (film): 3280 (w), 3090 (s), 3056 (sh), 3027 (m), 2924 (s), 2867 (sh), 1667 (m), 1604 (w), 1538 (s), 1491 (w), 1454 (w), 1410 (sh), 1390 (m), 1322 (s), 1236 (m), 1194 (w), 1070 (s), 1034 (sh), 1004 (sh), 965 (w), 923 (w), 878 (sh), 819 (s), 766 (w), 703 (s), 664 (w), 604 (w), 562 (w), 489 (s) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 7.16−7.14 (m, 3H, 2″,4″,6″-H, C_6H_5), 6.93−6.91 (m, 2H, 3″,5″-H, C_6H_5), 6.42 (d, ^3J = 3.2 Hz, 1H, 5-H), 5.87 (d, ^3J = 3.2 Hz, 1H, 4-H), 4.76 (s, 2H, OCH_2Fc), 4.31 (t, ^3J = 1.6 Hz, 2H, 2′,5′-H, Fc), 4.16 (s, 5H, C_5H_5, Fc), 4.13 (t, ^3J = 1.6 Hz, 2H, 3′,4′-H, Fc), 3.59 (s, 2H, SCH_2), 2.97 (s, 3H, NCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 150.9 (3-C), 138.9 (1″-C, C_6H_5), 128.8 (2″,6″-C, C_6H_5), 128.1 (3″,5″-C, C_6H_5), 126.6 (4″-C, C_6H_5), 121.4 (5-C), 105.8 (2-C), 96.3 (4-C), 83.4 (1′-C, Fc), 69.7 (2′,5′-C, Fc), 69.3 (OCH_2Fc), 68.5 (C_5H_5, Fc), 68.3 (3′,4′-C, Fc), 41.7 (SCH_2), 33.6 (NCH_3) ppm. The ^1H,^{13}C HMBC and ^1H,^{13}C HSQC 2D experiments provided additional support for the proposed structure. ^1H,^{15}N HMBC: δ −231.7 ppm. MS-EI (70 eV), *m/z* (%): 417 $[M]^{+\bullet}$ (4), 199 $[FcCH_2]^+$ (100), 197 (10), 183 (6), 121 $[C_5H_5Fe]^+$ (38), 91 (22), 56 $[Fe]^+$ (11).

*4.3.9.1. 2-(Benzylsulfanyl)−2-(ferrocenylmethyl)−1-methyl-1,2-dihydro-3H-pyrrol-3-one (**7i**) (Table 2).* ^1H NMR (400 MHz, $CDCl_3$): δ 7.39 (br s, 1H, 5-H), 7.21 (nr m, 5H, C_6H_5), 5.00 (br s, 1H, 4-H), 4.03 (s, 5H, C_5H_5, Fc), 3.98 (br s, 4H, 2′,3′,4′,5′-H, Fc), 3.62, 3.35 (both d, ^2J = 13.2 Hz, 2H, SCH_2), 3.12, 2.74 (both d, ^2J = 14.7 Hz, 2H, CH_2Fc), 2.88 (s, 3H, NCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.8 (*C* = *O*), 165.2 (5-C), 136.6 (1″-C, C_6H_5), 128.9, 128.2 (2″,3″,5″,6″-C, C_6H_5), 127.1 (4″-C, C_6H_5), 97.3 (4-C), 79.6 (1′-C, Fc), 76.8 (2-C), 69.9, 68.9 (2′,5′-C, Fc), 68.6 (C_5H_5, Fc), 68.0, 67.4 (3′,4′-C, Fc), 35.9 (*C*H_2Fc), 32.6 (NCH_3), 33.5 (SCH_2), ppm. ^1H,^{15}N HMBC: δ −269.4 ppm. MS-EI (70 eV), *m/z* (%): 417 $[M]^{+\bullet}$ (26), 295 (18), 248 (40), 230 (10), 221 (23), 228 (80), 199 $[FcCH_2]^+$ (100), 122 (13), 121 $[C_5H_5Fe]^+$ (48), 91 (50), 65 (12), 56 $[Fe]^+$ (22).

*4.3.10. Methyl {[3-(1-ferrocenylmethoxy)−1-methyl-1H-pyrrol-2-yl]sulfanyl}acetate (**2j**)*

*4.3.10.1. Methyl {[(Z)−2-(1-ferrocenylmethoxy)-N-methylbuta-2,3-dienimidoyl]sulfanyl}acetate (1-Aza-1,3,4-triene) (**5j**).* *n*-BuLi (7.5 mL of 2.37 M solution in hexane, 17.8 mmol) was added to a vigorously stirred solution of allene **1** (3.81 g, 15.0 mmol) in dry THF (25 mL) at −90 °C under argon. The temperature quickly rose to −50 °C. After stirring for an additional 10 min at −45 to −40 °C, the solution was cooled to −90 °C, and a mixture of MeNCS (1.10 g, 15.0 mmol) and THF (3 mL) was added in one portion. After stirring at −48 to −40 °C for 10 min, the mixture was cooled to −95 °C and sequentially were added a solution of H_2O (0.28 g, 15.5 mmol) in THF (2 mL), then (at −80 °C) $MeOC(O)CH_2Br$ (2.36 g, 15.4 mmol, 1.0 equiv). The cooling bath was removed. After raising the temperature to −30 °C (for 10 min), the reaction mixture was again cooled to −80 °C and quenched with a saturated aqueous NH_4Cl solution (10 mL). Usual workup gave a residue (6.17 g), dark-brown viscous liquid, consisting of 1-aza-1,3,4-triene **5j** (as a mixture of *E*- and *Z*-isomers) and unreacted allene **1** in a ratio of ~73:27 (by ^1H NMR spectroscopic analysis). IR (neat): 1945 (=*C*=) cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): (*major isomer*): δ 5.71 (s, 2H, CH_2=*C*=), 4.39 (s, 2H, OCH_2Fc), 4.23 (nr m, 2H, 2′,5′-H, Fc), 4.13 (nr m, 7H, 3′,4′-H, C_5H_5, Fc), 3.67 (s, 3H, CH_3OC=*O*), 3.66 (s, 2H, SCH_2), 3.29 (s, 3H, NCH_3) ppm; (*minor isomer*): δ 5.71 (s, 2H, CH_2=*C*=), 4.34 (s, 2H, OCH_2Fc), 4.25 (nr m, 2H, 2′,5′-H, Fc), 4.12 (nr m, 7H, 3′,4′-H, C_5H_5, Fc), 3.75 (s, 3H, CH_3OC=O), 3.71 (s, 2H, SCH_2), 3.33 (s, 3H, NCH_3) ppm; the ratio of isomers ~70:30. ^{13}C NMR (100 MHz, $CDCl_3$): (*major isomer*) δ 199.7 (=*C*=), 170.0 (C=O), 155.3 (N=C), 126.3 (OC=), 93.7 (CH_2=*C*=), 81.9 (1′-C, Fc), 70.0 (OCH_2Fc), 68.6 (3′,4′-C, Fc), 68.4 (C_5H_5, Fc), 67.2 (2′,5′-C, Fc), 52.2 (CH_3OC=*O*), 41.3 (NCH_3), 31.8 (SCH_2) ppm; (*minor isomer*): δ 200.2 (=C=), 169.1 (C=O), 157.3 (N=C), 130.1 (OC=), 92.7 (CH_2=*C*=), 81.3 (1′-C, Fc), 69.6 (OCH_2Fc), 68.4 (3′,4′-C, Fc), 67.8 (C_5H_5, Fc), 67.4 (2′,5′-C, Fc), 52.5 (*C*H_3OC=*O*), 40.9 (NCH_3), 33.6 (SCH_2) ppm.

*4.3.10.2. Cyclization of 1-aza-1,3,4-triene **5j** into 1H-pyrrole **2j**.* CuBr (0.15 g, 8.7 mol%) was added to a stirred solution of 1-aza-1,3,4-triene **5j** (6.02 g, 11.0 mmol; with taking into account of 27% impurity of allene **1**) in dry THF (20 mL) at r.t. The temperature rose to 43 °C. After stirring for an additional 30 min at r.t., the reaction mixture was treated with a saturated aqueous NH_4Cl solution (10 mL) contained NaCN (0.20 g). The layers were separated, and the products were extracted from the aqueous fraction with Et_2O (2 × 15 mL). The combined organic extracts were washed with H_2O (2 × 20 mL), dried ($MgSO_4$), passed through neutral alumina layer (2 cm), and concentrated under reduced pressure (on a rotary evaporator) to give a residue (5.23 g), dark-brown viscous liquid, consisting of pyrrole **2j**. Further purification by column chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1) gave 1.37 g (31%) pyrrole **2j** as yellow-orange needles, m.p. 98−99 °C; [Found: C, 57.29; H, 5.22; N, 3.65; S, 8.17. $C_{19}H_{21}FeNO_3S$ (399.386) requires C, 57.15; H, 5.30; N, 3.51; S, 8.03%]. IR (KBr): 3094, 2986, 2941, 2881, 1726, 1642, 1536, 1484, 1431, 1388, 1316, 1270, 1199, 1119, 1064, 1004, 959, 899, 820, 718, 661, 641, 600, 583, 506, 485, 420 cm^{-1}. ^1H NMR (400 MHz, $CDCl_3$): δ 6.60 (d, ^3J = 2.7 Hz, 1H, 5-H), 5.90 (d, ^3J = 2.7 Hz, 1H, 4-H), 4.80 (s, 2H, OCH_2Fc), 4.33 (nr m, 2H, 2′,5′-H, Fc), 4.20 (s, 5H, C_5H_5, Fc), 4.17 (nr m, 2H, 3′,4′-H, Fc), 3.64 (s, 3H, CH_3OC=*O*), 3.61 (s, 3H, NCH_3), 3.22 (s, 2H, SCH_2) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 170.4 (C=O), 151.4 (3-C), 122.2 (5-C), 104.2 (2-C), 96.0 (4-C), 83.1 (1′-C, Fc), 69.5 (2′,5′-C, Fc), 69.2 (OCH_2Fc), 68.4 (C_5H_5, Fc), 68.3 (3′,4′-C, Fc), 52.2 (*C*H_3OC=O), 38.7 (SCH_2), 34.2 (NCH_3) ppm. ^1H,^{15}N HMBC: δ −234.1 ppm.

*4.3.10.3. Methyl [(2-(ferrocenylmethyl)−1-methyl-3-oxo-2,3-dihydro-1H-pyrrol-2-yl)sulfanyl]acetate (**7j**) (Tentative assignment) (Table 2).* ^1H NMR (400 MHz, $CDCl_3$): δ 7.53 (br s, 1H, 5-H), 4.96 (br s, 1H, 4-H), 4.11 (br s, 2H, 2′,5′-H, Fc), 4.06 (s, 5H, C_5H_5, Fc), 3.97 (br s, 2H, 3′,4′-H, Fc), 3.66 (s, 3H, OCH_3), 3.18, 3.04 (both d, ^2J = 15.7 Hz, 2H, SCH_2), 3.14, 2.77 (both d, ^2J = 14.0 Hz, 2H, CH_2Fc), 3.06 (s, 3H, NCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.1 (C=O), 169.5 (CH_3OC=*O*), 165.7 (5-C), 97.2 (4-C), 79.2 (1′-C, Fc), 75.7 (2-C), 69.8, 68.7 (2′,5′-C, Fc), 68.6 (C_5H_5, Fc), 68.1, 67.4 (3′,4′-C, Fc), 52.2 (*C*H_3OC=*O*), 35.4 (OCH_2Fc), 32.7 (NCH_3), 30.9 (SCH_2) ppm. The ^1H,^{13}C HSQC 2D experiment provided additional support for the proposed structure.

*4.3.11. Tert–Butyl {[3-(1-ferrocenylmethoxy)−1-methyl-1H-pyrrol-2-yl]sulfanyl}acetate (**2k**)*

*4.3.11.1. tert–Butyl {[(Z)−2-(1-ferrocenylmethoxy)-N-methylbuta-2,3-dienimidoyl]sulfanyl}acetate (1-Aza-1,3,4-triene) (**5k**).* *n*-BuLi (7.5 mL of 2.37 M solution in hexane, 17.8 mmol) was added to a vigorously stirred solution of allene **1** (3.81 g, 15.0 mmol) in

dry THF (25 mL) at −90 °C under argon. The temperature quickly rose to −50 °C. After stirring for an additional 10 min at −45 to −40 °C, the solution was cooled to −90 °C, and a mixture of MeNCS (1.10 g, 15.0 mmol) and THF (3 mL) was added in one portion. After stirring at −48 to −40 °C for 10 min, the mixture was cooled to −95 °C and sequentially were added a solution of H_2O (0.28 g, 15.5 mmol) in THF (2 mL), then (at −80 °C) *t*-BuOC(O)CH_2Br (3.00 g, 15.3 mmol, 1.0 equiv). The cooling bath was removed. After raising the temperature to −30 °C (for 10 min), the reaction mixture was again cooled to −80 °C and quenched with a saturated aqueous NH_4Cl solution (10 mL). Usual workup gave a residue (6.86 g), brown oily liquid, consisting of 1-aza-1,3,4-triene **5k** (as a mixture of *E*- and *Z*-isomers) and unreacted allene **1** in a ratio of ~70:30 (by ^{1}H NMR spectroscopic analysis). IR (neat): 1948 (=C=) cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): (*major isomer*) δ 5.71 (s, 2H, CH_2=*C*=), 4.39 (s, 2H, OCH_2Fc), 4.24 (nr m, 2H, 2′,5′-H, Fc), 4.13 (s, 7H, 3′,4′-H, C_5H_5, Fc), 3.63 (s, 2H, SCH_2), 3.29 (s, 3H, NCH_3), 1.43 [s, 9H, $(CH_3)_3$COC=O] ppm; (*minor isomer*): δ 5.72 (s, 2H, CH_2=C=), 4.46 (s, 2H, OCH_2Fc), 4.27 (nr m, 2H, 2′,5′-H, Fc), 4.13 (s, 7H, 3′,4′-H, C_5H_5, Fc), 3.73 (s, 2H, SCH_2), 3.35 (s, 3H, NCH_3), 1.45 [s, 9H, $(CH_3)_3$COC=O] ppm; the ratio of isomers ~64:36. ^{13}C NMR (100 MHz, $CDCl_3$): (*major isomer*) δ 199.8 (=C=), 168.5 (C=O), 155.7 (N=C), 126.5 (OC=), 93.6 (*C*H_2=C=), 82.4 (1′-C, Fc), 82.0 [$(CH_3)_3$*C*OC=O], 69.0 (O*C*H_2Fc), 68.5 (3′,4′-C, Fc), 68.4 (C_5H_5, Fc), 67.2 (2′,5′-C, Fc), 41.2 (NCH_3), 33.5 (SCH_2), 27.9 [$(CH_3)_3$COC=O] ppm; (*minor isomer*): δ 200.5 (=*C*=), 167.8 (*C*=*O*), 157.8 (N=C), 130.5 (OC=), 92.8 (*C*H_2=C=), 81.5 (1′-C, Fc), 81.2 [$(CH_3)_3$*C*OC=*O*], 69.7 (O*C*H_2Fc), 68.7 (3′,4′-C, Fc), 68.4 (C_5H_5, Fc), 67.5 (2′,5′-C, Fc), 41.1 (NCH_3), 35.1 (SCH_2), 27.7 [(*C*H_3)$_3$COC=O] ppm.

*4.3.11.2. Cyclization of 1-aza-1,3,4-triene **5k** into 1H-pyrrole **2k**.* CuBr (0.15 g, 8.8 mol%) was added to a stirred solution of 1-aza-1,3,4-triene **5k** (6.80 g, 10.8 mmol; with taking into account of 30% impurity of allene **1**) in dry THF (20 mL) at r.t. The temperature rose to 43 °C. After stirring for an additional 30 min at r.t., the reaction mixture was treated with a saturated aqueous NH_4Cl solution (10 mL) contained NaCN (0.20 g). The layers were separated, and the products were extracted from the aqueous fraction with Et_2O (2 × 15 mL). The combined organic extracts were washed with H_2O (2 × 20 mL), dried ($MgSO_4$), passed through neutral alumina layer (2 cm), and concentrated under reduced pressure (on a rotary evaporator) to give a residue (5.90 g, 87%), dark-brown viscous liquid, consisting of pyrrole **2k**. Further purification by column chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1, 3:1) gave 0.96 g (20%) pyrrole **2k** as amber prisms, m.p. 64–65 °C; [Found: C, 59.77; H, 6.32; N, 3.09, S, 7.17. $C_{22}H_{27}FeNO_3S$ (441.366) requires C, 59.87; H, 6.17; N, 3.17; S, 7.27%]. IR (KBr): 3093, 2997, 2971, 2916, 2864, 1717, 1642, 1542, 1475, 1459, 1415, 1392, 1371, 1317, 1290, 1247, 1174, 1130, 1066, 1028, 1001, 963, 920, 826, 803, 758, 711, 654, 586, 500, 489, 428 cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.54 (d, ^{3} *J* = 3.1 Hz, 1H, 5-H), 5.85 (d, ^{3} *J* = 3.1 Hz, 1H, 4-H), 4.76 (s, 2H, OCH_2Fc), 4.28 (br t, ^{3} *J* = 1.7 Hz, 2H, 2′,5′-H, Fc), 4.16 (s, 5H, C_5H_5, Fc), 4.12 (br t, ^{3} *J* = 1.7 Hz, 2H, 3′,4′-H, Fc), 3.59 (s, 3H, NCH_3), 3.15 (s, 2H, SCH_2), 1.36 [s, 9H, $(CH_3)_3$COC=*O*] ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 169.2 (C=O), 151.1 (3-C), 121.8 (5-C), 104.8 (2-C), 96.1 (4-C), 83.3 (1′-C, Fc), 81.0 [$(CH_3)_3$COC=*O*], 69.6 (OCH_2Fc), 69.1 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.2 (3′,4′-C, Fc), 40.0 (SCH_2), 34.4 (NCH_3), 27.8 [$(CH_3)_3$COC=*O*] ppm. ^{1}H,^{15}N HMBC: δ −233.9 ppm.

*4.3.11.3. Tert–Butyl [(2-(ferrocenylmethyl)–1-methyl-3-oxo-2,3-dihydro-1H-pyrrol-2-yl)sulfanyl]acetate (**7k**) (Tentative assignment) (Table 2).* ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.52 (d, ^{3} *J* = 2.4 Hz, 1H, 5-H), 4.95 (d, ^{3} *J* = 2.4 Hz, 1H, 4-H), 4.11 (br s, 2H, 2′,5′-H, Fc), 4.06 (s, 5H, C_5H_5, Fc), 3.97 (br s, 2H, 3′,4′-H, Fc), 3.13, 3.10 (both d, ^{2} *J* = 8.3 Hz, 2H, SCH_2), 3.09 (s, 3H, NCH_3), 2.94, 2.77 (both d, ^{2} *J* = 14.1 Hz, 2H, CH_2Fc), 1.41 [s, 9H, $(CH_3)_3$COC=O] ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.8 (C=O), 168.1 [$(CH_3)_3$*C*OC=O], 165.6 (5-C), 97.1 (4-C), 81.9 [$(CH_3)_3$*C*OC=O], 79.4 (1′-C, Fc), 75.8 (2-C), 69.8, 68.7 (2′,5′-C, Fc), 68.6 (C_5H_5, Fc), 68.0, 67.4 (3′,4′-C, Fc), 35.5 (*C*H_2Fc), 32.8 (NCH_3), 32.5 (SCH_2), 27.8 [(*C*H_3)$_3$COC=O] ppm. ^{1}H,^{15}N HMBC: δ −233.8 ppm.

*4.3.12. {[3-(1-Ferrocenylmethoxy)–1-methyl-1H-pyrrol-2-yl]sulfanyl}acetonitrile (**2l**)*

To a vigorously stirred solution of allene **1** (2.54 g, 10.0 mmol) in dry THF (20 mL), *n*-BuLi (4.5 mL of 2.5 M solution in hexane, 11 mmol) was added at −90 °C under argon. After stirring for an additional 13 min at −46 to −38 °C, the solution was cooled to −80 °C, and a mixture of MeNCS (0.73 g, 10.0 mmol) and THF (3 mL) was added in one portion. After stirring at −43 to −40 °C for 10 min, the mixture was cooled to −95 °C and sequentially were added a solution of H_2O (0.20 g, 11 mmol) in THF (2 mL), then (at −83 °C) BrCH_2CN (1.36 g, 11 mmol, 1.1 equiv). After raising the temperature to −40 °C, the reaction mixture was again cooled to −60 °C and quenched with a saturated aqueous NH_4Cl solution (10 mL). Usual workup gave a residue (4.19 g), a dark-brown viscous liquid, consisting of 1-aza-1,3,4-triene **5l** (small amount), unreacted allene **1** and unidentified tar-like products (by ^{1}H NMR spectroscopic analysis).

To a stirred solution of the crude product (4.06 g) in THF (20 mL), CuBr (0.40 g) was added. After stirring for 40 min at 40–50 °C and usual workup, the reaction mixture was passed through neutral alumina layer (1.5 cm) and concentrated under reduced pressure (on a rotary evaporator) to give a residue (1.72 g, a dark-brown very viscous liquid) consisting of pyrrole **2l** (~22%) in a mixture with unreacted allene **1** (~78%). Further purification by column chromatography (neutral alumina, hexane, hexane/Et_2O, 1:1) and recrystallization (acetone/Et_2O) gave pyrrole 0.11 g (3%) **2l** as yellow rhombic crystals; m.p. 126–127 °C; [Found: C, 59.14; H, 4.78; N, 7.51; S, 8.66. $C_{18}H_{18}FeN_2OS$ (366.259) requires C, 59.03; H, 4.95; N, 7.65; S, 8.75%]. IR (KBr): 3122 (w), 3098 (w), 2965 (w), 2926 (w), 2869 (w), 1655 (w), 1532 (s), 1465 (w), 1386 (s), 1315 (s), 1237 (m), 1103 (w), 1068 (s), 1022 (sh), 1001 (m), 956 (m), 931 (sh), 856 (w), 817 (s), 791 (sh), 725 (s), 645 (w), 598 (w), 488 (s) cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.65 (d, ^{3} *J* = 3.2 Hz, 1H, 5-H), 5.91 (d, ^{3} *J* = 3.2 Hz, 1H, 4-H), 4.79 (s, 2H, OCH_2Fc), 4.26 (t, ^{3} *J* = 1.6 Hz, 2H, 2′,5′-H, Fc), 4.16 (s, 5H, C_5H_5, Fc), 4.14 (t, ^{3} *J* = 1.6 Hz, 2H, 3′,4′-H, Fc), 3.67 (s, 3H, NCH_3), 3.03 (s, 2H, SCH_2) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 152.4 (3-C), 124.1 (5-C), 117.4 (CN), 102.6 (2-C), 96.3 (4-C), 82.8 (1′-C, Fc), 69.9 (2′,5′-C, Fc), 69.5 (OCH_2Fc), 68.7 (3′,4′-C, Fc), 68.6 (C_5H_5, Fc), 34.6 (NCH_3), 22.7 (SCH_2) ppm. ^{1}H,^{15}N HMBC: δ −233.8 (pyrrole), −131.2 (CN) ppm.

*4.4. 3-(Allyloxy)–2-(methylsulfanyl)–1-phenyl-1H-pyrrole (**12a**)*

To a vigorously stirred solution of (allyloxy)allene (5.00 g, 52.0 mmol) in dry THF (70 mL), *n*-BuLi (32 mL of 1.6 M solution in hexane, 51.2 mmol) was added at −90 °C under argon. After stirring for an additional 10 min at ca. −70 to −65 °C, the solution was cooled to −100 °C and a mixture of PhNCS (6.76 g, 50.0 mmol) and THF (10 mL) was added in few portions. After being stirred for 10 min at −70 to −65 °C, the mixture was again cooled to ca. −80 °C and MeI (14.00 g, 98.6 mmol, 2 equiv) was added in one portion. The cooling bath was removed, and when the temperature was reached −10 °C, CuI (2.00 g) was added. The reaction mixture was stirred for 30 min at 40–45 °C, then cooled to r.t. and treated with saturated aqueous NH_4Cl solution (10 mL) contained NaCN (~2.00 g). The layers were separated, and the aqueous phase was extracted with Et_2O (4 × 30 mL). The organic layers were combined, washed with saturated aqueous NH_4Cl solu-

tion (60 mL), dried ($MgSO_4$). Concentration under reduced pressure gave a residue (12.10 g), dark mobile liquid, consisting of corresponding pyrrole **12a** (by ^{1}H NMR). Pure compound **12a** was isolated by column chromatography on neutral alumina, using hexane and hexane/Et_2O (10:1) as eluent. Yield: 5.49 g (45%); light-yellow oily liquid, n_D^{24} 1.6005; [Found: C, 68.39; H, 6.09; N, 5.69; S, 13.02. $C_{14}H_{15}NOS$ (245.341) requires C, 68.54; H, 6.16; N, 5.71; S, 13.07%]. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.39–7.27 (m, 5H, C_6H_5), 6.72 (d, ^3J = 3.3 Hz, 1H, 5-H), 6.06 (ddt, $^3J_{trans}$ = 17.4 Hz, $^3J_{cis}$ = 10.6 Hz, ^3J = 5.3 Hz, 1H, C*H*=CH_2), 6.00 (d, ^3J = 3.3 Hz, 1H, 4-H), 5.38 (ddd, $^3J_{trans}$ = 17.4 Hz, $^4J \approx {}^2J$ = 1.3 Hz, 1H, CH=CH_2), 5.22 (ddd, $^3J_{cis}$ = 10.6 Hz, $^4J \approx {}^2J$ = 1.3 Hz, 1H, CH=CH_2), 4.55 (br d, ^3J = 5.3 Hz, 2H, OCH_2), 1.99 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 151.0 (3-C), 139.6 (1′-C, C_6H_5), 133.9 (CH=CH_2), 128.5 (3′,5′-C, C_6H_5), 126.8 (4′-C, C_6H_5), 125.9 (2′,6′-C, C_6H_5), 121.5 (5-C), 117.1 (CH=CH_2), 107.9 (2-C), 97.6 (4-C), 71.8 (OCH_2), 20.0 (SCH_3) ppm.

4.5. 3-(Benzyloxy)–2-(methylsulfanyl)–1-phenyl-1H-pyrrole (**12b**)

To a vigorously stirred solution of (benzyloxy)allene (2.47 g, 16.9 mmol) in dry THF (25 mL), *n*-BuLi (6 mL of 2.5 M solution in hexane, 15.0 mmol) was added at −93 °C under argon. After stirring for an additional 5 min at ca. −52 to −45 °C, the solution was cooled to −80 °C and a mixture of PhNCS (2.03 g, 15.0 mmol) and THF (5 mL) was added in one portion. After being stirred for 5 min at −52 to −42 °C, the mixture was again cooled to ca. −80 °C and MeI (4.30 g, 30.3 mmol, 2.0 equiv) was added in one portion. The cooling bath was removed, and when the temperature was reached −5 °C, CuI (0.60 g) was added. The reaction mixture was stirred for 35 min at 33–45 °C, then cooled to r.t. and treated with saturated aqueous NH_4Cl solution (15 mL) contained NaCN (0.50 g). The layers were separated, and the aqueous phase was extracted with Et_2O (15 mL). The organic layers were combined, washed with water (20 mL), and dried ($MgSO_4$). Concentration under reduced pressure gave a residue (4.70 g, a brown oily liquid), from which pyrrole **12b** was isolated by column chromatography (neutral alumina, hexane, hexane/Et_2O, 10:1) in yield of 50% (2.22 g) as light-yellow crystals; m.p. 48–49 °C; [Found: C, 73.10; H, 5.75; N, 4.69; S, 10.66. $C_{18}H_{17}NOS$ (295.400) requires C, 73.19; H, 5.80; N, 4.74; S, 10.86%]. IR (KBr): 3152 (w), 3101 (w), 3062 (w), 3035 (w), 2988 (w), 2919 (w), 1629 (w), 1595 (m), 1537 (s), 1497 (s), 1450 (m), 1395 (m), 1348 (s), 1315 (sh), 1225 (w), 1199 (w), 1165 (w), 1135 (w), 1084 (sh), 1067 (s), 1020 (sh), 993 (m), 938 (w), 911 (w), 764 (m), 733 (s), 694 (s), 640 (w), 568 (w), 469 (w) cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.46–7.24 (m, 10H, $OCH_2C_6H_5$, NC_6H_5), 6.70 (d, ^3J = 3.3 Hz, 1H, 5-H), 5.98 (d, ^3J = 3.3 Hz, 1H, 4-H), 5.09 (s, 2H, $OCH_2C_6H_5$), 1.98 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 151.2 (3-C), 139.7 (1′-C, NC_6H_5), 137.8 (1′-C, $OCH_2C_6H_5$), 128.6 (3′,5′-C, NC_6H_5), 128.3 (3′,5′-C, $OCH_2C_6H_5$), 127.6 (4′-C, NC_6H_5), 127.3 (2′,6′-C, $OCH_2C_6H_5$), 127.0 (4′-C, $OCH_2C_6H_5$), 126.1 (2′,6′-C, NC_6H_5), 121.6 (5-C), 108.3 (2-C), 98.0 (4-C), 73.0 ($OCH_2C_6H_5$), 20.2 (SCH_3) ppm. ^{1}H,^{15}N HMBC: δ −211.7 ppm. MS-EI (70 eV), *m/z* (%): 295 $[M]^{+\bullet}$ (12), 204 (100), 176 (8), 170 (33), 143 (12), 135 (9), 104 (12), 91 (57), 77 (36), 65 (15), 51 (15).

4.6. 3-(Ferrocenylmethoxy)–2-(methylsulfanyl)–1-vinyl-1H-pyrrole (**17**)

To a stirred solution of *t*-BuOK (0.29 g, 2.6 mmol) in DMSO (3 mL), *N*-[2-(*vinyloxy*)*ethyl*]pyrrole **2c** (0.40 g, 1.0 mmol) was added. After stirring at r.t. for 50 min, the reaction mixture was diluted with water (10 mL) and products were extracted with Et_2O (3 × 5 mL). The combined organic solution was washed with water (3 × 5 mL), dried ($MgSO_4$), and concentrated under reduced pressure to give a residue (0.35 g, solid) from which *N*-vinylpyrrole **17** (0.338 g, 96%) was isolated by recrystallization from hexane as a yellow-brown crystals, m.p. 86–87 °C; [Found: C, 61.08; H, 5.33; N, 4.05; S, 8.93. $C_{18}H_{19}FeNOS$ (353.26) requires C, 61.20; H, 5.42; N, 3.96; S, 9.08%]. IR (KBr): 3126 (w), 3098 (w), 2983 (w), 2917 (w), 2876 (w), 1637 (s), 1550 (s), 1478 (w), 1409 (sh), 1374 (s), 1302 (w), 1263 (w), 1244 (sh), 1203 (sh), 1104 (w), 1068 (m), 1007 (m), 943 (m), 869 (w), 815 (m), 724 (m), 658 (w), 586 (w), 510 (s), 488 (s) cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.29 (dd, $^3J_{trans}$ = 16.0 Hz, $^3J_{cis}$ = 8.9 Hz, 1H, NC*H*=CH_2), 6.97 (d, ^3J = 3.4 Hz, 1H, 5-H), 6.03 (d, ^3J = 3.4 Hz, 1H, 4-H), 4.80 (s, 2H, OCH_2Fc), 5.07, 4.66 (both dd, $^3J_{trans}$ = 16.0 Hz, $^3J_{cis}$ = 8.9 Hz, $^2J_{gem}$ = 1.0 Hz, 2H, NCH=CH_2), 4.28 (t, ^3J = 1.7 Hz, 2H, 2′,5′-H, Fc), 4.16 (s, 5H, C_5H_5, Fc), 4.13 (t, ^3J = 1.7 Hz, 2H, 3′,4′-H, Fc), 2.09 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 150.9 (3-C), 130.3 (NCH=CH_2), 116.2 (5-C), 108.3 (2-C), 100.1 (4-C), 96.6 (NCH=*C*H_2), 83.0 (1′-C, Fc), 70.0 (OCH_2Fc), 69.3 (2′,5′-C, Fc), 68.5 (C_5H_5, Fc), 68.4 (3′,4′-C, Fc), 20.5 (SCH_3) ppm. ^{1}H,^{15}N HMBC: δ −213.9 ppm. MS-EI (70 eV), *m/z* (%): 353 $[M]^{+\bullet}$ (9), 200 (25), 199 $[FcCH_2]^+$ (100), 197 (12), 121 $[C_5H_5Fe]^+$ (42), 56 $[Fe]^+$ (13).

Monitoring of the reaction showed that the quantitative transformation of pyrrole **2c** into *N*-vinylpyrrole **17** takes place after 30 min.

4.6.1. 2-(Ferrocenylmethyl)–2-(methylsulfanyl)–1-vinyl-1,2-dihydro-3H-pyrrol-3-one (**18**) (Tentative assignment) (Scheme 8)

^{1}H NMR (400 MHz, $CDCl_3$): δ 7.89 (d, ^3J = 3.7 Hz, 1H, 5-H), 6.38 (dd, $^3J_{trans}$ = 15.8 Hz, $^3J_{cis}$ = 9.2 Hz, 1H, NC*H*=CH_2), 5.18 (d, ^3J = 3.7 Hz, 1H, 4-H), 4.79, 4.50 (both d, $^3J_{trans}$ = 15.8 Hz, $^3J_{cis}$ = 9.2 Hz, 2H, NCH=CH_2), 4.07, 4.04 (br s, 2H, 3′,4′-H, Fc), 4.05 (s, 5H, C_5H_5, Fc), 3.97, 3.95 (br s, 2H, 2′,5′-H, Fc), 3.07, 2.92 (both d, ^2J = 14.0 Hz, 2H, CH_2Fc), 1.77 (s, 3H, SCH_3) ppm. ^{13}C NMR (100 MHz, $CDCl_3$): δ 199.5 (C=O), 158.2 (5-C), 129.6 (NCH=CH_2), 102.2 (4-C), 93.5 (NCH=CH_2), 79.2 (1′-C, Fc), 75.5 (2-C), 70.1, 69.4 (2′,5′-C, Fc), 68.6 (C_5H_5, Fc), 67.9, 67.6 (3′,4′-C, Fc), 36.1 (CH_2Fc), 11.2 (SCH_3) ppm. ^{1}H,^{15}N HMBC: δ −246.8 ppm.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jorganchem.2020.121651.

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